Nothing Beats Zero, and the Standard Tests Cannot Tell: Simulation Calibration of Forecast Comparison on Correlated Equity Panels

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Abstract

Short-horizon equity forecasting studies report a model that beats a benchmark. This one reports the apparatus, calibrates it against known truth, and finds three of its standard components badly wrong. On synthetic panels whose predictable share is fixed by construction, the two-sided Diebold–Mariano test applied to panel rows has size 0.23 at four names correlated at 0.57 and 0.79 at a hundred, against a nominal 0.05; the inflation is $2\Phi(-z_{1-\alpha/2}/\sqrt{1+(k-1)\bar{\rho}})$ in closed form, which the simulation tracks to within 0.007 across the sweep, and Holm's family-wise error reaches 0.59. Against a zero-forecast benchmark the models are nested, and Diebold–Mariano declares a pure-noise fitted forecast significantly *worse* than the benchmark 92% of the time. The False Strategy Theorem threshold is cleared by half of all skill-free grids at every level of dependence between trials, because it is an expectation and not a critical value. Three replacements are calibrated and hold their nominal level: aggregating loss differentials to one value per session, the Clark–West statistic, and a stationary bootstrap that resamples an entire configuration grid on a single index draw. Applied to a leakage-controlled Borsa Istanbul panel with an executable next-open-to-close target, the corrected stack finds no predictive content in any of six fitted forecasters, and an exact joint p-value of 0.77 for the best of 72 configurations, measured to behave like 3.8 independent trials. The uncorrected stack would have reported four models as significantly worse than the null—an artefact of fitting, not a finding. Inverting the corrected tests, the design could only have resolved an out-of-sample R^2 above 0.11, against the 0.01 this literature treats as meaningful. The negative result is therefore a statement about the experiment rather than about the market.

1 Introduction

A short-horizon equity forecasting result is easy to manufacture and hard to verify. The standard failure modes are well catalogued: a target the backtest could not have traded, a validation split that lets a model see its own future, a metric whose null is never stated, a p-value computed on rows that are not independent, and a headline drawn from whichever configuration happened to look best. Each inflates apparent performance without leaving an obvious trace in the code.

There is a second failure mode, less discussed, that applies with equal force to the studies that report nothing. A negative result carries information only in proportion to the power of the design that produced it. "We tested and found no effect" and "we ran a test that could not have found the effect even had it been there" are different claims, and in this literature they are almost never distinguished, because the second requires computing something the first does not. Gelman and Carlin [1] make the general argument: a design should be characterised by what it can resolve before its output is interpreted. Button et al. [2] show what happens to a field that does not do this.

There is a third failure mode, and it is the subject of this paper. Both of the arguments above assume the tests themselves work. A p-value is a claim about a long-run frequency, and that claim is a theorem only under conditions the data are supposed to satisfy: independent observations, non-nested models, trials that disperse like independent draws. Panel forecast comparison satisfies none of them. Four tickers observed on one session are not four observations. A fitted forecast compared against a zero forecast is nested inside it. Seventy-two configurations run over overlapping folds are not seventy-two searches. Each violation is well known in isolation; what is not known, and what nobody appears to have measured, is how far the resulting numbers are from the numbers they claim to be, on data with the dependence structure these studies actually have.

We measure it. Synthetic panels are generated whose cross-sectional correlation, volatility and predictable share are fixed by construction and matched to a real panel, the production estimators are run on them unmodified, and the rate at which each one is wrong is recorded. The answers are not marginal. The two-sided equal-accuracy test applied to panel rows rejects a true null 23% of the time on a four-name panel and 79% on a hundred-name panel. Holm's family-wise error rate reaches 59%. Against a zero benchmark the squared-error comparison declares a forecast that is pure estimation noise significantly *worse* than the benchmark in 92% of replications, and in all of them once the record is long enough. The False Strategy Theorem's threshold, widely reported as though it were a critical value, is cleared by about half of all skill-free grids, which is exactly what an expectation should do.

None of these is a curiosity of a badly behaved corner. The first has a closed form the simulation reproduces to three decimal places, so it can be evaluated before an experiment is run. The second is a consequence of nesting that grows with the fitted model's own variance and therefore penalises exactly the models a study is most interested in. The third is a definitional error that survives because the quantity it produces looks like a threshold.

The rest of the paper does three things with this. It calibrates a replacement for each broken component and shows the replacements hold their level under heavy tails, clustered volatility, regime switches, and record lengths from 60 to 500 sessions. It applies the corrected apparatus to a leakage-controlled Borsa Istanbul panel, where the verdict is negative but—importantly—negative in a different way than the uncorrected apparatus would have reported. And it inverts the corrected tests to ask what the design could have detected, which turns the negative result from a verdict about the market into a verdict about the experiment.

Contributions

C1. A measured error budget for the standard panel-forecast evaluation stack. Section 5 reports the size of each test in the usual pipeline on panels whose truth is fixed by construction and whose dependence is matched to a real one: the equal-accuracy test at row level, the same test under Holm correction, the Reality Check, both variants of the test of superior predictive ability, and the False Strategy threshold. Three of them are wrong by margins large enough to reverse published conclusions, and the errors are reported with Monte Carlo intervals rather than asserted.

C2. A closed form for the dependence inflation, and its verification (Proposition 1). The limiting size of a pooled test on k units equicorrelated at $\bar{\rho}$ is $2\Phi(-z_{1-\alpha/2}/\sqrt{1+(k-1)\bar{\rho}})$. The simulation lands on this curve at every width tested, the largest discrepancy across a size range from 0.11 to 0.79 being 0.0064. A practitioner can therefore evaluate the damage before running anything, and a reader can evaluate it for a published study from two numbers that study already reports.

C3. Three calibrated replacements. Aggregating loss differentials to one value per session restores the nominal level at every cross-section width and under every stress applied (Section 5.2); this is Fama and MacBeth [3] applied to loss differentials rather than to returns. The Clark–West statistic [4] removes the nesting bias and holds its level where the squared-error comparison reaches 92% (Section 5.3). A stationary bootstrap that resamples every configuration on one index draw supplies an exact p-value and a critical value for the best of a correlated grid, where the deflated Sharpe ratio supplies neither (Section 5.5).

C4. Detectability bounds for the design. Inverting the accepted tests at conventional size and power yields the smallest out-of-sample R^2 and the smallest Sharpe ratio the experiment could have separated from its null (Section 6), by simulation as well as in closed form. Both exceed anything the literature credibly reports, by an order of magnitude.

C5. An information ceiling for correlated panels (Proposition 2). Under equicorrelation, one session of a k-name cross-section carries $k/(1 + (k - 1)\bar{\rho})$ independent observations, which is bounded above by $1/\bar{\rho}$. Breadth therefore cannot substitute for length: past a few dozen names, adding more buys essentially no statistical precision. On this panel an unlimited universe would improve standard errors by 9%.

C6. A breadth–cost feasibility bound (Proposition 3). For a long-only rule holding the top k of N ranked names against a round-trip cost c and a target of volatility σ, positive expected net return requires a cross-sectional information coefficient above $c/(\sigma\,\lambda(N, k))$, where λ is the mean of the top-k standard normal order statistics. The bound is a property of the experimental design, computable before any model is fitted, and this design misses it by a factor of thirty. Its two breadth regimes differ (Proposition 4): at a fixed holding fraction the requirement converges, while at a fixed holding count it falls to zero, so the bound reports the universe a design would need rather than declaring any design impossible.

C7. An effective trial count for correlated searches (Proposition 5), measured rather than assumed. The False Strategy Theorem assumes independent trials; a configuration grid reuses most of the same sessions and disperses less. Two independent routes to the number of searches the grid really represents are reported: matching expected maxima in closed form gives 7.01 of 72, and the joint bootstrap—which needs no independence assumption at all—gives 3.77, at a measured mean pairwise correlation of 0.839 between configurations. The simulation recovers the nominal count when the trials genuinely are independent, which is the check that licenses reading either number.

C8. A verified evaluator. The inference code is itself subject to mutation testing: twenty real defects are reintroduced one at a time and each is required to break a specific test (Appendix F). Four tests were found to be decorative this way and repaired. The committed run replays byte-for-byte from bundled inputs, and every table and figure in this manuscript is generated from that bundle, or from the calibration study's own hashed artifact, by a tested module.

The empirical outcome is negative. It is retained because the calibration is what makes it interpretable, and because the same calibration shows that the uncorrected apparatus would have reported a different negative result—one that a reader could reasonably have mistaken for a finding.

2 Related work

Forecast comparison. Diebold and Mariano [5] give the standard test of equal predictive accuracy; Harvey et al. [6] show the statistic is oversized in small samples and supply the correction used here. Diebold [7] later emphasised that the test compares forecasts rather than the models producing them, which is the interpretation adopted below, and warned specifically against applying the test to nested comparisons—a warning Section 5.3 quantifies. Giacomini and White [8] reframe the comparison as one of conditional predictive ability under a fixed estimation window, which sidesteps the nesting problem by changing the null rather than by correcting the statistic.

Nested models. When one forecast nests another, the population loss differential is zero under the null and the numerator of the Diebold–Mariano statistic degenerates; Clark and McCracken [9] and McCracken [10] derive the resulting non-standard limiting distributions. Clark and West [4] give the adjustment used here, which restores approximate normality by removing the term the fitted model's own estimation variance contributes. This literature is mature and unambiguous, and it is nonetheless routine for applied short-horizon studies to compare a fitted model against a zero or historical-mean forecast with an unadjusted squared-error test. Section 5.3 reports what that costs: not a mild size distortion but a near-certain verdict in the benchmark's favour, whatever the data contain.

Cross-sectional dependence in panels. Petersen [11] shows that standard errors in finance panels are badly understated when a common time effect is ignored, and that the size of the error is a function of the correlation and the cross-section's width; Thompson [12] gives two-way clustered formulas. Fama and MacBeth [3] is the older remedy and the one adopted here: reduce each period's cross-section to a single number and conduct inference on the resulting series. Section 5.2 shows that applying this to loss differentials rather than to returns restores the nominal level of a forecast comparison exactly, and gives the closed form for the error it removes.

Data snooping. White [13] shows that selecting the best of m models and testing that model alone is invalid, and gives a bootstrap Reality Check for the joint null. Hansen [14] studentizes the statistic and recentres the bootstrap; Romano and Wolf [15] extend the idea to a stepwise procedure that identifies which models survive, and Hansen et al. [16] to a confidence set over models. Sullivan et al. [17] apply the Reality Check to a century of technical trading rules and find the apparent performance of the best rule largely disappears once the size of the search is accounted for; that result is the direct template for Section 7.2.

Dependent-data resampling. Politis and Romano [18] introduce the stationary bootstrap; Politis and White [19], corrected by Patton et al. [20], give the automatic block-length rule used here.

Backtest overfitting. Lo [21] derives the sampling distribution of the Sharpe ratio and shows the square-root annualisation rule is wrong under autocorrelation. Bailey and López de Prado [22] convert the Sharpe ratio into a probability accounting for skewness and kurtosis, and Bailey and López de Prado [23] deflate it by the number of configurations tried. Bailey et al. [24] show how quickly an unrecorded search produces a spurious backtest, and Harvey et al. [25] make the same argument for the cross-section of expected returns. What none of this literature supplies, and what Section 6.4 adds, is a way to say how many *independent* searches a correlated grid amounts to.

Effect sizes worth detecting. Welch and Goyal [26] document that most equity-premium predictors fail out of sample; Campbell and Thompson [27] argue that an out-of-sample R^2 of well under one percent is already economically meaningful, which is the scale against which Section 6 measures this design. Gu et al. [28] report out-of-sample R^2 in the same range for machine-learning models on a monthly US cross-section far larger than this one. Any short-horizon study claiming an R^2 an order of magnitude above that range is more likely reporting a leak than a discovery.

Design analysis. Cohen [29] is the standard reference for inverting a test at a stated power. Gelman and Carlin [1] argue that this should be done with a plausible effect size supplied externally rather than with the observed estimate, which is the procedure of Section 6; Button et al. [2] and Ioannidis [30] document the consequences of skipping it. To our knowledge this is not standard practice in the empirical finance forecasting literature, where negative results are reported without any accompanying statement of what the design could have resolved.

Calibrating an inference procedure by simulation. The practice of generating data from a known truth and checking that a procedure recovers it at its advertised rate is standard in computational statistics and rare in empirical finance. Cook et al. [31] formalise it for Bayesian software by checking the uniformity of posterior quantiles, and Talts et al. [32] generalise the construction. Section 5 is the frequentist analogue for a forecast-evaluation pipeline: the object being validated is not a posterior sampler but the chain of estimators that turns saved predictions into p-values, and the criterion is not quantile uniformity but the measured size of each test against its nominal level. We are not aware of a published study that reports this for the equity forecast-comparison stack, which is the gap the paper is addressed to.

Breadth and costs. Grinold [33] relates the information ratio of an active strategy to the product of skill and the square root of breadth. Proposition 3 is a cost-aware statement in the same spirit but with the opposite use: rather than predicting attainable performance from assumed skill, it converts an observed cost schedule into the skill a design requires, using the order-statistic machinery of David and Nagaraja [34]. Novy-Marx and Velikov [35] document that many published anomalies do not survive realistic trading costs; Section 7.3 reproduces that pattern in miniature, and Section 6.3 explains why it was inevitable here.

Borsa Istanbul. Kara et al. [36] report high directional accuracy for neural networks and support vector machines on the Istanbul Stock Exchange index. That study, like most in the genre, reports directional accuracy at index level without transaction costs, without a stated null, and without a correction for the number of specifications examined. The present work does not contradict it; it measures a harder quantity on a different universe and reports what survives when those three things are supplied.

3 Data

The committed run contains 1,004 provider rows for GARAN, ISCTR, KCHOL and THYAO from 2025-04-07 to 2026-04-03. Every execution open and every volume observation is flagged `observed`; proxy opens are rejected, because a proxy open cannot be traded at. Each record carries provider name, provider symbol, source record identifier, retrieval timestamp, and separate open- and volume-quality flags, and the canonical record keeps raw tradable, split-adjusted and total-return price representations separately.

The runner validates expected sessions, duplicates, weekend rows, holidays, timezone, and full-day and half-day session bounds against the committed Borsa Istanbul schedule. The input bundle carries five sourced cash-dividend events and a per-ticker corporate-action query-coverage artifact, so "no action" is an assertion with a source rather than an absence of data. Because the accepted strategy holds for one session and carries no overnight entitlement, those five events are persisted as `no_entitlement` rather than invented income. Split, bonus-issue, ticker-change, rights-issue and delisting handlers exist and are exercised by synthetic tests; the rights-issue and unsourced-delisting paths fail closed rather than guess a policy.

The universe is fixed by construction and therefore free of survivorship bias within its own definition. It is *not* point-in-time index membership, and no claim about BIST-100 is made anywhere in this document.

4 Method

4.1 Executable target

For ticker i and feature date t, only information available after the official close of session t is used. With O and C the raw tradable open and close,

$$r_{i,t+1}^{\text{exec}} = \frac{C_{i,t+1}}{O_{i,t+1}} - 1, \qquad y_{i,t+1} = \mathbf{1}\left\{r_{i,t+1}^{\text{exec}} > 0\right\}. \tag{1}$$

Every panel row stores the feature, execution and target timestamps, and the chronology $t_{\text{feature}} < t_{\text{exec}} \leq t_{\text{start}} < t_{\text{end}}$ is enforced as an invariant rather than assumed.

4.2 Features

Nineteen features enter the pooled model, listed in full in Appendix B: returns over one, five and twenty sessions, moving-average and VWAP ratios, an ATR-to-close ratio, realised volatility, a volume z-score, a drawdown, two cross-sectional quantities, and four calendar encodings. Every one is a ratio, a log ratio, a rank or a bounded periodic function.

Raw close, moving-average levels, absolute ATR, raw VWAP, raw volume and OBV are deliberately excluded. In a pooled model, nominal price scale acts as accidental ticker identification: a model that learns "prices near 300" has learned an identity, not a signal, and the leak is invisible in any per-ticker diagnostic. Missing values are preserved with a reason code and imputed by training-fold median with an accompanying indicator column, so absence is a feature rather than a silently filled zero.

4.3 Validation protocol

Folds are expanding windows over unique trading dates. A training date group is purged when any of its samples has a feature timestamp or target end reaching the first validation feature timestamp, so for every fold k,

$$\max_{j \in \text{train}(k)} t_j^{\text{end}} < \min_{j \in \text{val}(k)} t_j^{\text{feature}}. \tag{2}$$

Tickers on one date are never split across partitions, and preprocessing is fitted on training rows only. The committed run uses 24 minimum training dates, 10 validation dates, a 10-date step and a one-date embargo, producing 12 folds over 120 disjoint evaluation sessions. Figure 1 draws the partition that was executed, read from the run's own fold artifact; Appendix D tabulates its boundaries.

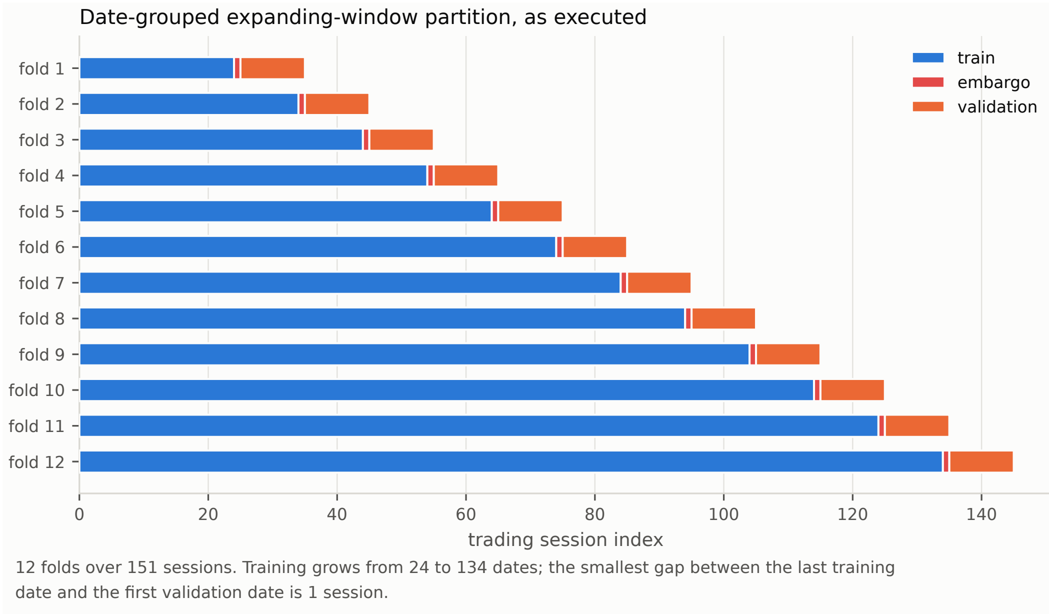


Figure 1: The executed date-grouped expanding-window partition. Nothing is illustrative: each bar spans the dates the fold really used.

Table 1: Cross-sectional dependence in the evaluation panel, and the most an arbitrarily wide cross-section could add at the same correlation.

Quantity	Value
Panel units	4
Evaluated sessions	120
Out-of-sample rows	480
Mean within-session correlation	0.5697
Variance inflation factor	2.7090
Effective independent rows	177.2
Mean within-session correlation of the loss differential	0.5662
Independent rows per session, from the loss differential	1.4823
Ceiling on independent rows per session ($1/\bar{\rho}$)	1.7663
Standard-error gain from an unlimited universe	9.2%

4.4 Effective sample size and the unit of inference

For k cross-sectional units with average pairwise correlation $\bar{\rho}$, the variance of a cross-sectional mean is inflated by

$$\mathrm{VIF} = 1 + (k-1)\bar{\rho}, \qquad n_{\mathrm{eff}} = \frac{n}{\mathrm{VIF}}. \tag{3}$$

Rather than apply a correction after the fact, the evaluation changes the unit of inference: every loss differential is averaged across the tickers present on a date and all tests run on the resulting one-value-per-session series. Row-level statistics are computed anyway and reported alongside, so the size of the inflation is visible rather than argued about. The estimator averages pairwise Pearson correlations with equal weight over unit pairs, skipping pairs with fewer than three shared sessions rather than treating them as zero, and floors the inflation factor at one: a negative average correlation would imply the pooled sample carries *more* information than independent rows, which is not a claim this evaluation is willing to make. Table 1 and Figure 2 give the realised values.

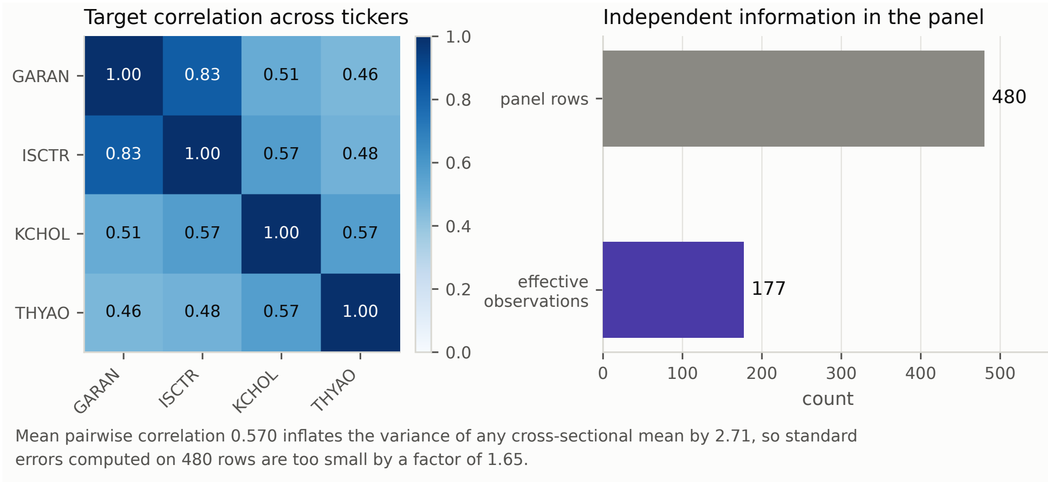


Figure 2: Realised correlation of the executable target across the four tickers, and the sample size it implies.

4.5 Estimators and metrics

Seven forecasters share the same panel, target, folds, feature manifest, evaluation dates and train-only preprocessing: a zero-return null, a training-fold majority-direction null, a one-session persistence rule, a same-date cross-sectional mean, a rolling training mean, a regularised logistic direction model, and a ridge return model that also ranks the portfolio. The three history heuristics see only targets that had already matured at the prediction timestamp, which is the leakage guard that makes them baselines rather than oracles.

The primary accuracy metric is the zero-mean out-of-sample R^2,

$$R_0^2 = 1 - \frac{\sum_{i,t}(r_{i,t} - \hat{r}_{i,t})^2}{\sum_{i,t} r_{i,t}^2}. \tag{4}$$

The sample-mean version is the wrong choice here: its benchmark uses the mean of the evaluation window, which is not available at prediction time, so it flatters any model that merely learns the window's drift.

4.6 Tests

Equal predictive accuracy. With session-aggregated squared-error differentials $d_t = L_t(\text{model}) - L_t(\text{null})$,

$$DM = \frac{\bar{d}}{\sqrt{\hat{\Omega}/n}}, \qquad DM^* = DM\sqrt{\frac{n+1-2h+h(h-1)/n}{n}}, \tag{5}$$

with $\hat{\Omega}$ the Bartlett long-run variance and $h = 1$, referred to a Student t distribution on $n-1$ degrees of freedom [5, 6]. The sign convention is reported with every test: a *positive* statistic means the model is worse than the null. Six models are tested against the same null on the same data, so Holm's step-down procedure [37] controls the family-wise error rate without assuming independence.

Data snooping. With $Z_{k,t} = L_t(\text{null}) - L_t(\text{model } k)$,

$$V_n = \max_k \sqrt{n}\,\bar{Z}_k, \qquad T_n = \max\left[0,\ \max_k \frac{\sqrt{n}\,\bar{Z}_k}{\hat{\omega}_k}\right], \tag{6}$$

whose null distributions come from a stationary bootstrap that resamples every model on the same index draw, preserving cross-model dependence, with the block length chosen automatically [13, 14, 18, 19]. One deviation from Hansen [14] is recorded here because it matters in exactly this regime: the p-values compare untruncated maxima rather than the floored statistic T_n. Applying the floor to both the observed value and every bootstrap draw collapses the comparison onto the atom at zero as soon as no candidate has a positive mean, and returns an arbitrary p-value precisely when the evidence is weakest. Because this is a judgement call rather than a derivation, both quantities are reported and the size of both is measured: Table 5 puts the variant at 0.0755 and Hansen's own definition at 0.0765 on the anchor design, a difference well inside Monte Carlo error, and neither exceeds 0.083 anywhere in the sweep. The deviation is therefore immaterial to size, and the untruncated form is preferred only because it remains informative in the region where the floored form does not.

Sharpe ratio under search. For per-period Sharpe $\hat{SR}$ with skewness $\hat{\gamma}_3$ and kurtosis $\hat{\gamma}_4$,

$$\widehat{PSR}(SR^*) = \Phi\left[\frac{(\hat{SR} - SR^*)\sqrt{n-1}}{\sqrt{1 - \hat{\gamma}_3 \hat{SR} + \frac{\hat{\gamma}_4 - 1}{4}\hat{SR}^2}}\right], \tag{7}$$

and the deflated Sharpe ratio evaluates it at the threshold the best of N skill-free trials would be expected to reach,

$$SR_0^* = \sqrt{V[\hat{SR}]}\, q(N), \qquad q(N) = (1-\gamma)\,\Phi^{-1}\big(1 - \tfrac{1}{N}\big) + \gamma\,\Phi^{-1}\big(1 - \tfrac{1}{Ne}\big), \tag{8}$$

with γ the Euler–Mascheroni constant [22, 23]. Both quantities are per period; substituting an annualised value inflates the result silently. Lo's autocorrelation-aware annualisation factor [21] is reported alongside the square-root rule, with lags truncated at the Newey–West bandwidth [38, 39] because 120 sessions cannot estimate 251 autocorrelations.

4.7 Portfolio simulation

Long-only, top-k by predicted return, equal weights, next-open execution at observed prices, one-session holding. A name is bought only when its predicted return net of the declared decision-cost assumption is positive. Participation caps and market impact use a 20-session trailing volume reference whose last observation is no later than the signal date. The ledger records signals, orders, fills, positions, cash and costs as separate artifacts and enforces $E_T = E_0 + \mathrm{PnL}_{\mathrm{gross}} + \mathrm{distributions} - \mathrm{costs}$ session by session. Cost sensitivity reuses the same trading decisions across multipliers, so only the bill moves. The complete parameter set is in Appendix C.

5 Calibrating the evaluation stack

Everything in Section 4 is standard practice. This section asks whether standard practice works, by running the same estimators on data whose answer is known before they see it.

5.1 The generator

A panel of k units over n sessions is assembled from four orthogonal unit-variance channels: a common predictable factor z^p, a common unpredictable factor z^u, and their idiosyncratic counterparts e_i^p and e_i^u. With $\bar{\rho}$ the cross-sectional correlation, θ the share of return variance that is in principle predictable, and ν_t a unit mean-square volatility path shared by every name,

$$r_{i,t} = \sigma\,\nu_t\Big[\sqrt{\bar{\rho}\theta}\,z_t^p + \sqrt{(1-\bar{\rho})\theta}\,e_{i,t}^p + \sqrt{\bar{\rho}(1-\theta)}\,z_t^u + \sqrt{(1-\bar{\rho})(1-\theta)}\,e_{i,t}^u\Big]. \tag{9}$$

Pairwise correlation is then exactly $\bar{\rho}$ whatever the volatility path does, and the predictable component carries exactly a share θ of the variance. Building the panel from named channels rather than from a single draw is what lets a forecast be specified by its *population* moments: a forecast is a linear combination of the two predictable channels and two fresh noise channels, so its variance, its cross-sectional correlation and its covariance with the target are set rather than estimated afterwards. In particular the population out-of-sample R_0^2 of a forecast is

$$R_0^2 = \frac{2\,\mathrm{cov}(r,\hat{r}) - \mathbb{E}[\hat{r}^2]}{\mathbb{E}[r^2]}, \tag{10}$$

which is zero at a covariance ratio equal to the forecast's own variance ratio, positive above it and negative below. That identity is the reason the two tests in this paper answer different questions, and it is the axis every power curve below is drawn against.

Table 2: Measured size of the two-sided equal-accuracy test at a nominal 5%, against the closed form of Proposition 1, as the cross-section widens at the panel's measured correlation. Brackets are 95% Monte Carlo intervals over 10,000 replications. Treating rows as independent is not a small approximation; aggregating to the session removes the error entirely.

Names k	Predicted (row)	Measured (row)	Measured (session)
2	0.1127	0.1121 [0.1060, 0.1184]	0.0459 [0.0419, 0.0502]
4	0.2220	0.2284 [0.2202, 0.2368]	0.0529 [0.0486, 0.0575]
10	0.4128	0.4185 [0.4088, 0.4282]	0.0461 [0.0421, 0.0504]
30	0.6270	0.6281 [0.6185, 0.6376]	0.0492 [0.0450, 0.0536]
100	0.7878	0.7913 [0.7832, 0.7992]	0.0456 [0.0416, 0.0499]

The volatility path is Gaussian and constant in the baseline and is replaced in turn by a GARCH(1,1) recursion, a two-state switching level, Student t_5 innovations, and both together. Every path is rescaled to mean square one, so a heteroskedastic cell differs from a constant one in its clustering and not in its scale.

The design point the empirical chapter is entitled to cite is fixed by measurement rather than by preference: four names, 120 evaluated sessions, target correlation 0.5697, target volatility 0.0190, and a forecast whose variance ratio 0.272 and cross-sectional correlation 0.947 are those of the ridge model in the committed run. Only the predictable share cannot be measured, and it is set to the value that reproduces the measured variance ratio at a population accuracy of exactly zero.

5.2 The size of a test applied to panel rows

Proposition 1 (Size of a pooled test under equicorrelation)**.** *Let a session contain k units whose loss differentials are equicorrelated at $\bar{\rho}$, and let a two-sided test at nominal level α divide the pooled mean by $\sqrt{\hat{\sigma}^2/(kT)}$, ignoring the dependence. Its rejection probability under the null converges to*

$$\alpha^*(k, \bar{\rho}) = 2\,\Phi\left(-\frac{z_{1-\alpha/2}}{\sqrt{1+(k-1)\bar{\rho}}}\right), \tag{11}$$

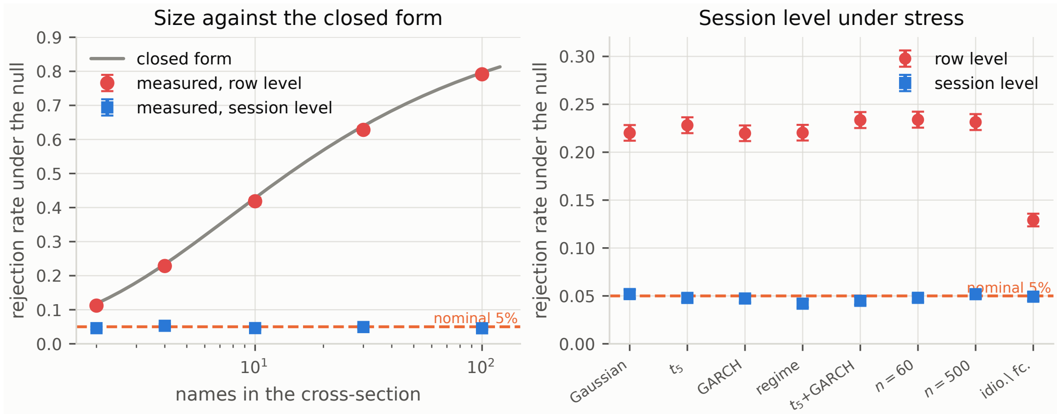
which is increasing in both arguments and tends to one as $k \to \infty$ for any $\bar{\rho} > 0$.

The proof is two lines and is given in Appendix A. Its value is not its difficulty but its falsifiability: it makes a point prediction for every cell of a simulation that was written independently of it.

Table 2 and the left panel of Figure 3 report the comparison. At the measured correlation of 0.5697 the row-level test rejects a true null 22.84% of the time on four names against a predicted 22.20%, 62.81% on thirty against a predicted 62.70%, and 79.13% on a hundred against a predicted 78.78%. The largest discrepancy anywhere in the sweep is 0.0064, which is the order of the Monte Carlo error at ten thousand replications. Equation 11 is therefore not an approximation to be checked but a description to be used.

Two features of the table deserve emphasis because they are easy to misread. First, the failure is not a small-sample effect: it does not shrink as the record lengthens, because lengthening the record does not make the rows within a session any less dependent. Table 3 confirms this directly, with the row-level size at 0.2314 over 500 sessions against 0.2339 over 60. Second, the failure is not confined to wide panels. Even two names inflate a 5% test to 11%, and it is common for short-horizon studies to pool a handful of tickers without remark.

Aggregating to one differential per session removes the error completely. The session-level column of Table 2 never leaves $[0.0456, 0.0529]$ across a hundred-fold change in the cross-section's width, and Table 3 holds it inside $[0.0419, 0.0519]$ under heavy tails, clustered volatility, a switching level, both together, and record lengths from 60 to 500 sessions. The single case that drifts below nominal is the regime-switching generator at 0.0419, which is conservative rather than liberal. This is the Fama and MacBeth [3] construction applied to loss differentials, and its cost is the loss of within-session information that Proposition 2 shows is largely illusory in the first place.



Left: the size of the equal-accuracy test applied to panel rows is a closed-form function of the cross-section's width and correlation, and the simulation lands on it at every width. Right: the session-aggregated test holds its nominal level under heavy tails, clustered volatility, a switching level, and record lengths from 60 to 500 sessions.

Figure 3: Left: the measured size of the row-level equal-accuracy test against the closed form of Proposition 1. Right: the session-aggregated test under five departures from the baseline generator and two record lengths.

Table 3: Measured size of the session-aggregated test at a nominal 5% under departures from the baseline generator. The aggregation is what holds the level; none of the stresses that break the row-level test disturbs it.

Generator	Row level	Session level
Gaussian, constant volatility	0.2201 [0.2120, 0.2284]	0.0519 [0.0476, 0.0564]
Student t_5 innovations	0.2281 [0.2199, 0.2365]	0.0479 [0.0438, 0.0523]
GARCH(1,1) volatility	0.2197 [0.2116, 0.2279]	0.0473 [0.0432, 0.0516]
Regime-switching level	0.2203 [0.2122, 0.2286]	0.0419 [0.0381, 0.0460]
Student t_5 with GARCH	0.2335 [0.2252, 0.2419]	0.0449 [0.0409, 0.0491]
60 sessions	0.2339 [0.2256, 0.2423]	0.0480 [0.0439, 0.0524]
500 sessions	0.2314 [0.2232, 0.2398]	0.0518 [0.0475, 0.0563]
Idiosyncratic forecast	0.1291 [0.1226, 0.1358]	0.0492 [0.0450, 0.0536]

5.3 What a squared-error comparison does to a nested model

Comparing a fitted forecast against a zero forecast under squared-error loss asks a question the fitted model cannot win fairly. Writing $d_t = \hat{r}_t^2 - 2r_t\hat{r}_t$ for the differential against a zero benchmark,

$$\mathbb{E}[d] = \mathbb{E}[\hat{r}^2] - 2\,\mathrm{cov}(r, \hat{r}), \tag{12}$$

so a forecast carrying genuine but small covariance with the target still loses whenever its estimation variance exceeds twice that covariance. Under the null that the *population* forecast is zero, the fitted forecast is not zero: it is noise, with a variance the estimation procedure put there. The test therefore rejects towards the benchmark by construction, and the rate at which it does so is a property of the fitting procedure rather than of the market.

Table 4 and Figure 4 measure that rate. At the variance ratio the committed run's ridge model actually exhibits, the session-aggregated Diebold–Mariano test declares the fitted model significantly worse than the benchmark in 92.19% of replications in which the benchmark is exactly correct. At a variance ratio of 0.60 it reaches 99.88%, and over 500 sessions it reaches 100%: more data makes the error more certain, not less, because the bias is in the estimand and not in the standard error. Only the low-variance cell, at a ratio of 0.05, falls to 30.91%, which is still six times the nominal level. The complementary rate—the test declaring the fitted model significantly *better*—never exceeds 0.04%.

Table 4: The nested null: the zero benchmark is the correct population forecast and the fitted model is estimation noise. The squared-error comparison is not merely oversized, it is almost certain to declare the fitted model significantly *worse*, at a rate set by the forecast's own variance. The Clark–West statistic, which removes that term, holds its level.

Design	$\mathbb{E}[\hat{y}^2]/\mathbb{E}[r^2]$	DM rejects towards null	Clark–West
Anchor design	0.272	0.9219 [0.9165, 0.9271]	0.0508 [0.0466, 0.0553]
Low-variance forecast	0.050	0.3091 [0.3000, 0.3183]	0.0534 [0.0491, 0.0580]
High-variance forecast	0.600	0.9988 [0.9979, 0.9994]	0.0513 [0.0471, 0.0558]
Uncorrelated names	0.272	1.0000 [0.9996, 1.0000]	0.0532 [0.0489, 0.0578]
30 names	0.272	0.9545 [0.9502, 0.9585]	0.0565 [0.0521, 0.0612]
500 sessions	0.272	1.0000 [0.9996, 1.0000]	0.0521 [0.0478, 0.0566]

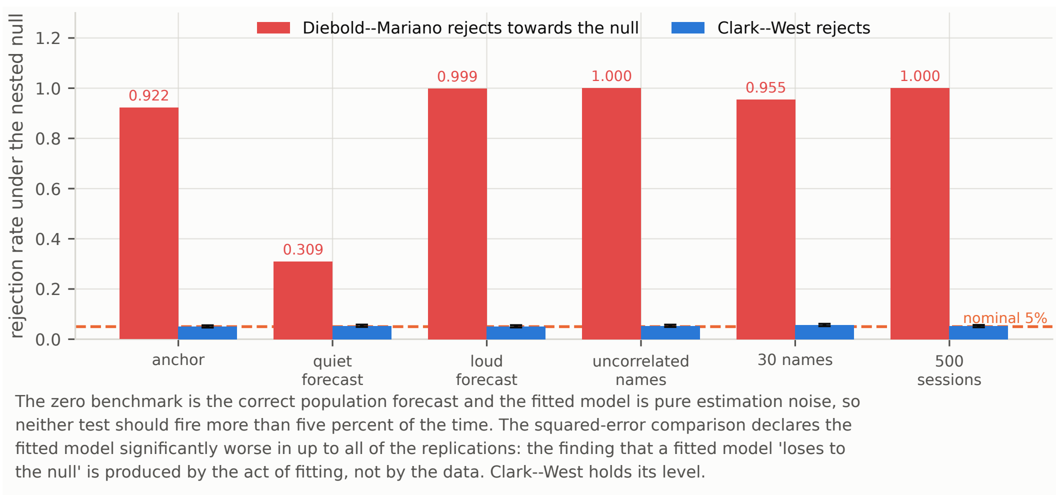


Figure 4: Under the nested null the zero benchmark is the correct population forecast and the fitted model is estimation noise. The squared-error comparison convicts the fitted model almost always; the Clark–West statistic holds its level.

The consequence for how such a result should be read is direct, and it applies to this paper's own earlier reporting as much as to anyone else's. A study that compares fitted models against a zero or historical-mean benchmark under squared-error loss and reports that several of them are "significantly worse than the benchmark" has, with high probability, reported the output of Equation 12 rather than a fact about its data. The finding is consistent with the models being worthless, and equally consistent with their carrying real information that is smaller than their own estimation variance. The two cannot be distinguished by that test.

Clark and West [4] remove the offending term. Adding back $(\hat{r}_t - 0)^2$ gives $f_t = 2r_t\hat{r}_t$, whose expectation is zero under the null and positive whenever the forecast covaries with the target, however weakly. The resulting one-sided test is correctly sized at every cell of Table 4, between 0.0508 and 0.0565, including the cells where the squared-error comparison is at 100%. Its row-level counterpart is not: at thirty names it reaches 0.3496, so the aggregation of Section 5.2 is required for this test as well.

The two tests answer different questions and this study reports both. Diebold–Mariano asks whether the forecast would have been worth using; Clark–West asks whether it contains predictive information at all. A forecast can fail the first and pass the second, and that combination—real information, too small to pay for the variance it costs—is the single most likely state of affairs in short-horizon equity prediction.

Table 5: Family-wise error rate at a nominal 5% with every member of a family of 6 forecasts skill-free by construction, so any rejection anywhere in the family is a false one. Holm controls the family but not the dependence: applied to row-level p-values on a correlated panel it fails outright, while the same procedure on session-aggregated p-values is correct. The bootstrap corrections are mildly liberal throughout. Over 2,000 replications the widest 95% Monte Carlo half-width in the table is 0.022.

Correction	k=4 corr.	k=4 indep.	k=30 corr.	k=30 indep.
Uncorrected	0.185	0.198	0.184	0.184
Holm, row level	0.225	0.042	0.588	0.035
Holm, session level	0.042	0.043	0.038	0.035
White Reality Check	0.059	0.057	0.056	0.051
SPA, untruncated	0.075	0.070	0.082	0.063
SPA, Hansen	0.076	0.070	0.083	0.064

5.4 Family-wise error and the multiple-comparison layer

Table 5 and Figure 5 repeat the exercise for a family of six forecasts, every member skill-free by construction, so any rejection anywhere in the family is a false one.

Holm's procedure controls the family-wise error rate without assuming independence between the hypotheses, and it does. What it cannot do is repair p-values that were wrong before it saw them: applied to row-level statistics on a correlated panel its family-wise error is 0.2250 at four names and 0.5875 at thirty, while the identical procedure applied to session-aggregated statistics gives 0.0420 and 0.0385. The correction is not the problem; its inputs are.

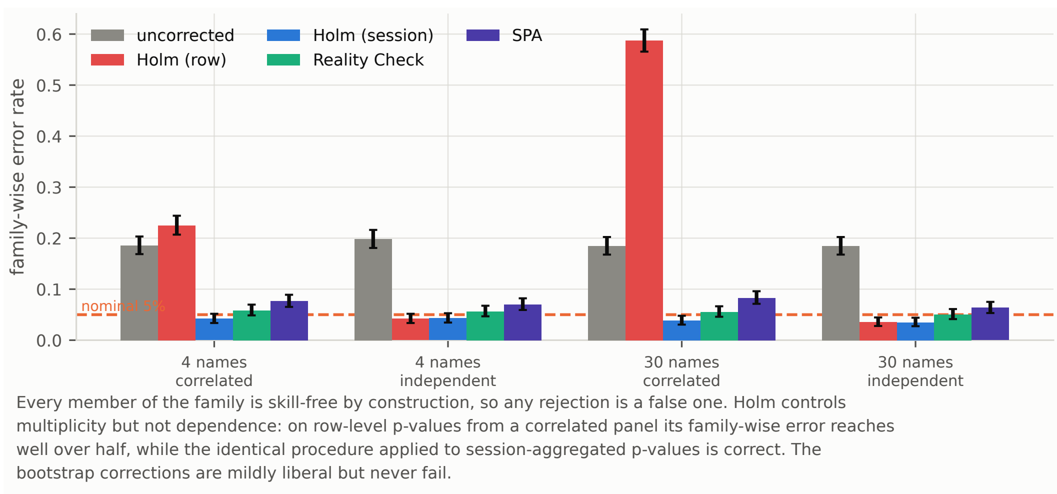


Figure 5: Family-wise error with every family member skill-free by construction. Holm applied to row-level p-values on a correlated panel fails; the same procedure on session-aggregated p-values is correct.

The bootstrap corrections behave well without the aggregation, because the stationary bootstrap resamples sessions and therefore carries the within-session dependence with them. The Reality Check sits between 0.0505 and 0.0585; the two variants of the test of superior predictive ability sit between 0.0630 and 0.0830, mildly liberal in every cell and most so on the widest correlated panel. That is worth stating plainly rather than burying: a nominal 5% SPA test on a design like this one is closer to an 8% test.

5.5 What a correlated search is worth

The deflated Sharpe ratio corrects the best of N trials using the False Strategy Theorem's expected maximum, $\sqrt{V}\, q(N)$ with q as in Equation 8. Two things can go wrong. The trials may not disperse

Table 6: A grid of 72 skill-free configurations, at four levels of dependence between them. The False Strategy expectation is exceeded about half the time whatever the dependence, because an expectation is not a critical value; the synchronised bootstrap quantile of the same grid holds the nominal level. The last column is the independent-trial count the grid behaves like, recovered from the simulated maximum.

Trial corr.	Clears $\mathbb{E}[\max]$	Clears bootstrap $q_{0.95}$	N_{eff}
0.00	0.457 [0.426, 0.488]	0.056 [0.043, 0.072]	67.10
0.50	0.500 [0.469, 0.531]	0.044 [0.032, 0.059]	12.95
0.90	0.484 [0.453, 0.515]	0.046 [0.034, 0.061]	2.66
0.98	0.490 [0.459, 0.521]	0.046 [0.034, 0.061]	1.67

like independent draws, which Proposition 5 addresses by solving for an effective count. And the threshold is an *expectation*, which a skill-free grid exceeds about half the time by definition.

Table 6 and Figure 6 measure both. Across trial correlations from 0 to 0.98, a grid of 72 skill-free configurations clears the expected maximum in 45.7% to 50.0% of replications. The rate is essentially flat in the dependence, which is the signature of the definitional error rather than of a dependence problem: whatever the correlation, an expectation is cleared about half the time. Reporting a grid maximum below this threshold is therefore a weak statement, and reporting one above it is not evidence of anything at all.

Resampling the entire grid on a single stationary-bootstrap index draw fixes this. The draw preserves both the serial dependence within a configuration and the cross-sectional dependence between configurations, and its 95th percentile is a critical value rather than a mean. Its measured size is 0.0560, 0.0440, 0.0460 and 0.0460 at the four correlation levels: correct throughout, and notably correct at $\bar{\rho} = 0.98$, where the analytic correction has no purchase at all.

The same simulation validates the effective-trial diagnostic. When the trials are genuinely independent it recovers 67.10 of a nominal 72, which is the check that the construction is not simply reporting a small number whatever it is given; at correlations of 0.5, 0.9 and 0.98 it returns 12.95, 2.66 and 1.67.

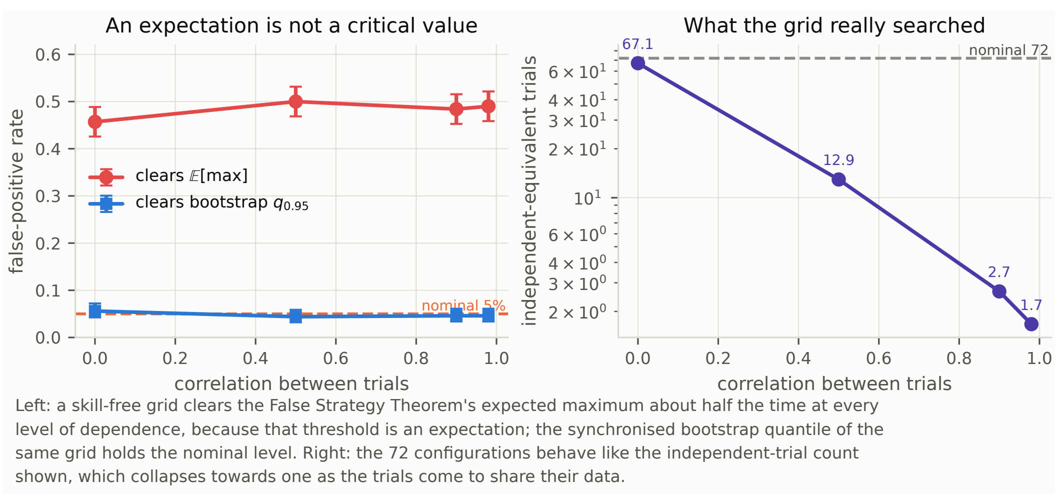


Figure 6: Left: a skill-free grid clears the False Strategy expectation about half the time at every level of dependence; the synchronised bootstrap quantile holds its level. Right: the independent-trial count the grid behaves like, recovered from the simulated maximum.

5.6 Detectable effects, by simulation

Table 7 inverts the two tests by simulation rather than by a normal approximation: at each record length the population effect is raised until the measured rejection rate reaches 80%. At 120 sessions

Table 7: The smallest effect each test separates from its null at 80% power and 5% size, obtained by simulating the whole comparison rather than by inverting a normal approximation. The Diebold–Mariano column is a population out-of-sample R_0^2; the Clark–West column is a population covariance ratio, which is the quantity that test responds to. A forecast can carry information the second detects while losing the comparison the first makes.

Sessions	DM: R_0^2 at 80% power	Clark–West: covariance ratio
120	0.2012	0.1912
250	0.1395	0.1526
500	0.1146	0.0957
1,000	0.0712	0.0769
2,000	0.0625	0.0473

the equal-accuracy test requires a population R_0^2 of 0.2012 and Clark–West requires a covariance ratio of 0.1912. Both fall roughly as $n^{-1/2}$ thereafter, reaching 0.0625 and 0.0473 at two thousand sessions.

These numbers are larger than the closed-form inversion of Section 6 reports, and the gap is informative rather than contradictory. The closed form uses central t quantiles and treats the standard error as scaling exactly as $n^{-1/2}$; the simulation makes neither assumption and pays for the difference. Where the two disagree the simulation is the one to believe, and the direction of the disagreement means the closed form is optimistic about what such designs can resolve.

5.7 What is and is not established here

The calibration is a statement about estimators applied to data with a stated dependence structure, not about markets. Its force comes from the fact that the structure is measured from a real panel rather than chosen for convenience, and that the estimators are the production ones: the fast path used to form loss differentials is checked against the pandas path the empirical chapter runs, to floating-point tolerance, by a test that fails if they diverge.

Three limits are worth stating. The generator is equicorrelated, so it does not reproduce a factor structure with heterogeneous loadings; the closed form of Proposition 1 is correspondingly an average-case statement. Forecast families are generated by adding independent noise to a shared predictable component, which reproduces the correlation structure of a real model family only approximately. And the search experiment imposes an exchangeable correlation between trials, whereas a real configuration grid has neighbours that are more alike than distant points. In every case the direction of the approximation is towards understating the problem, so the measured error rates should be read as lower bounds on what the corresponding real designs suffer.

6 What the design could have detected

Everything above answers “is there evidence of skill?”. This section answers the question those tests cannot: would they have said yes for an effect of the size anyone actually expects? Each quantity below is computed from the run’s own artifacts, and each is reported in Table 18, Table 19 or Table 20.

6.1 Minimum detectable accuracy

The statistic in Equation 5 is a mean divided by its standard error. Inverting it at size α and power $1-\beta$ gives the smallest mean loss differential the design could have separated from zero,

$$\delta_{\min} = \big(t_{1-\alpha/2,\,n-1} + t_{1-\beta,\,n-1}\big) \cdot \mathrm{SE}(\bar{d}), \tag{13}$$

and dividing by the null’s mean squared error puts it on the scale of Equation 4. We evaluate this for the candidate with the *smallest* standard error, the one the design had the best chance of separating; quoting any other would flatter the experiment. Following Gelman and Carlin [1] the reference effect is supplied externally rather than taken from the data: we use $R_0^2 = 0.01$, at the upper end of what Campbell and Thompson [27] and Gu et al. [28] treat as economically meaningful out of sample.

Since SE $\propto n^{-1/2}$, the session count at which a given effect becomes detectable follows by solving Equation 13 for n. The critical values move with n as well, so the solution is found by search rather than by the closed form $n_0(\delta_0/\delta)^2$, which understates the requirement at small n.

The same inversion applies to the portfolio. Setting Equation 7 equal to 0.95 at the search threshold of Equation 8 and solving for $\hat{SR}$ gives the per-session Sharpe ratio that would have counted as evidence of skill after the search actually performed.

Remark 1. Whether SR_0^* shrinks with the sample size depends entirely on what $V[\hat{SR}]$ stands for, and the distinction is easy to lose. Under Lo's iid expression $V = (1 + \hat{SR}^2/2)/n$ the threshold falls like $n^{-1/2}$, so a longer record does lower the bar the search sets. The threshold is a floor that a longer record cannot erode only if V measures persistent heterogeneity in the *true* Sharpe ratios of the configurations. The realised cross-trial variance this run plugs in is neither: it mixes true heterogeneity, estimation noise, and the fact that the trials cover different evaluation windows. We therefore report the threshold at the sample size actually observed and do not extrapolate it, and we make no claim that a strategy below it could never establish skill at any record length.

6.2 The information ceiling of a correlated panel

Widening the cross-section is the usual answer to a short sample. It does not work when the units share a market factor.

Proposition 2 (Panel information ceiling)**.** *Let a session contain k units whose targets are equicorrelated at $\bar{\rho} \in (0, 1)$. Then the session carries*

$$m(k) = \frac{k}{1 + (k-1)\bar{\rho}} \tag{14}$$

independent observations, m is strictly increasing in k, and $m(k) \uparrow 1/\bar{\rho}$ as $k \to \infty$.

The proof is immediate from Equation 3 and is given in Appendix A. The consequence is the part that matters: the gain available from breadth is bounded by $1/(\bar{\rho}\, m(k))$ in variance and by its square root in standard error, so a study whose cross-section is already moderately wide gains almost nothing from widening it further. Only more sessions help. Figure 14 draws the curve and marks where this panel sits on it.

6.3 The cost floor on breadth

Breadth does buy something else: selectivity. Ranking N names and holding the best k concentrates the forecast into its right tail, and the strength of that concentration is the mean of the top-k standard normal order statistics [34],

$$\lambda(N, k) = \frac{1}{k} \sum_{i=N-k+1}^{N} \mathbb{E}\big[Z_{(i:N)}\big] = \frac{N}{k} \int_0^1 \Phi^{-1}(u)\, I_u(N-k,\, k)\, \mathrm{d}u, \tag{15}$$

with I_u the regularised incomplete beta function. The second form follows from exchangeability—a draw is in the top k exactly when at most $k-1$ of the other $N-1$ exceed it—and replaces k quadratures by one.

Proposition 3 (Breadth–cost feasibility)**.** *Let the forecast $\hat{r}$ and the realised return $r \sim \mathcal{N}(0, \sigma^2)$ be jointly normal with correlation ρ. A long-only rule that ranks N names by $\hat{r}$, holds the top k in equal weight for one period, and pays a round-trip cost c on notional, has positive expected net return only if*

$$\rho \; > \; \frac{c}{\sigma\, \lambda(N, k)}. \tag{16}$$

The proof is in Appendix A. Four features of the bound are worth stating plainly. It is *generous*: it ignores estimation error in the ranking, charges the cost only once per round trip, and assumes an unbiased forecast. It is *model-free*: c, σ and $\lambda(N, k)$ are properties of the experimental design, not of the forecaster, so the requirement can be computed before a single model is fitted. It is *not* a statistical statement: a strategy can fail Equation 16 while producing a forecast that is significantly better than the null, which is exactly what happens when a real but small edge is smaller than the spread.

And it is *not an impossibility result*, which is the reading it most invites. The two breadth regimes diverge. At a fixed holding fraction $q = k/N$, $\lambda \to \varphi(\Phi^{-1}(1-q))/q$, a finite tail mean, so the requirement converges and extra names stop helping. At a fixed holding count, $\lambda(N,1) \sim \sqrt{2\log N}$ grows without bound, so the requirement falls to zero and a sufficiently wide ranking always satisfies the bound. What the proposition delivers is therefore the universe a design would need, not a proof that none suffices: and in the fixed-count regime it omits precisely the capacity, participation and impact constraints that bind hardest on a rule holding one name out of many. Proposition 4 states both limits.

Proposition 4 (Breadth regimes)**.** *With λ as in Equation 15, for fixed $q \in (0,1)$, $\lambda(N, \lceil qN \rceil) \to \varphi(\Phi^{-1}(1-q))/q < \infty$ as $N \to \infty$; while for fixed k, $\lambda(N,k) \to \infty$, with $\lambda(N,1) \sim \sqrt{2\log N}$.*

Remark 2. Proposition 3 selects within a session, so ρ is the *per-session cross-sectional* correlation between forecast and realised return. A correlation pooled over stock-sessions is a different quantity: it also absorbs the common time-series component, so a forecast that only tracks the market direction earns pooled correlation while ranking nothing. Both are reported in Table 19, and they differ here by a factor of four.

Equation 16 is a cost-aware relative of the fundamental law of active management [33], used in the opposite direction: rather than predicting attainable performance from assumed skill, it converts an observed cost schedule into the skill a design demands.

6.4 How many searches was the grid?

Equation 8 assumes the N trials are independent draws. A configuration grid is not: neighbouring configurations reuse most of the same sessions, so their Sharpe ratios move together and disperse less than independent trials would. Substituting the *realised* dispersion V_{real} absorbs the dependence but hides its strength.

Proposition 5 (Effective trial count)**.** *Let $V_{\text{ind}} = (1 + \hat{SR}^2/2)/n$ be the sampling variance of a single per-period Sharpe ratio [21], and let V_{real} be the realised variance of the N trial Sharpe ratios. Then q in Equation 8 is continuous and strictly increasing on $(1,\infty)$, so*

$$\sqrt{V_{\text{real}}}\, q(N) = \sqrt{V_{\text{ind}}}\, q(N_{\text{eff}}) \tag{17}$$

has a unique solution N_{eff}, the number of independent searches whose expected maximum equals the grid's.

When $V_{\text{real}} < V_{\text{ind}}$ the grid is behaving like fewer independent searches than it contains, and $N_{\text{eff}} < N$. Reporting both thresholds brackets the correction: the independent-trial reading is the conservative one, the realised-dispersion reading is the permissive one, and a grid maximum below both is a conclusion that does not depend on the choice.

7 Results

7.1 No fitted model reaches the null

Every fitted model has negative out-of-sample R^2 (Table 8 and Figure 7). Under Holm correction, four of the six are significantly *worse* than the null and none is better (Table 9); `logistic` and `rolling_mean` are indistinguishable from it.

That sentence is the one the calibration forces us to unlearn, and we state the correction plainly because the reading it displaces is the natural one and we had adopted it ourselves. That reading—several models are reliably worse than predicting zero, which is what fitting to a signal-free panel produces—treats the four rejections as information. They are not. Section 5.3 shows that this exact comparison, at this exact forecast variance ratio, produces significant rejections towards the benchmark in 92.19% of replications in which the benchmark is exactly correct. Four of six is not evidence

Table 8: Out-of-sample accuracy by model, recomputed from the saved prediction rows. The zero-mean R^2 compares against a zero forecast, so the null sits at exactly zero by construction.

Model	n	MAE	RMSE	R_0^2	Spearman IC	Dir. acc.
`zero_return`	480	1.425%	1.898%	0.0000	n/a	53.12%
`rolling_mean`	480	1.461%	1.962%	-0.0680	0.0079	50.42%
`logistic`	480	1.502%	1.971%	-0.0783	-0.0040	48.75%
`ridge`	480	1.651%	2.120%	-0.2470	0.0210	51.04%
`market_direction`	480	1.873%	2.458%	-0.6768	0.0094	52.71%
`majority_direction`	480	1.978%	2.466%	-0.6876	0.1080	53.12%
`previous_return`	480	1.984%	2.634%	-0.9252	0.0324	51.25%

Table 9: Equal predictive accuracy against the zero-return null under squared-error loss. The differential is $\mathrm{loss(model)} - \mathrm{loss(null)}$, so a positive statistic means the model is worse. The final column repeats each test treating all panel rows as independent draws.

Model	DM^*	p	Holm p	Verdict	Row-level p
`majority_direction`	+5.334	0.0000	0.0000	loses to null	0.0000
`previous_return`	+4.707	0.0000	0.0000	loses to null	0.0000
`market_direction`	+3.824	0.0002	0.0008	loses to null	0.0000
`ridge`	+3.334	0.0011	0.0034	loses to null	0.0000
`logistic`	+1.945	0.0541	0.1081	indistinguishable	0.0015
`rolling_mean`	+1.642	0.1031	0.1081	indistinguishable	0.0103

Table 10: Clark–West tests of predictive content against the zero-return benchmark. The adjusted differential is $f_t = 2r_t\hat{r}_t$, whose expectation is zero under the null that the population forecast is zero and positive whenever the forecast covaries with the target. The test is one-sided and, unlike the squared-error comparison in Table 9, is correctly sized for this nested pair.

Model	$\bar{f}$	CW	p	Holm p	Verdict
`ridge`	+1.664e-05	+0.621	0.2672	1.0000	none
`previous_return`	+2.937e-05	+0.596	0.2757	1.0000	none
`majority_direction`	+1.698e-05	+0.366	0.3572	1.0000	none
`logistic`	+3.795e-06	+0.268	0.3944	1.0000	none
`market_direction`	+3.482e-06	+0.077	0.4695	1.0000	none
`rolling_mean`	-3.275e-06	-0.229	0.5904	1.0000	none

that the models are bad; it is slightly fewer rejections than the null hypothesis itself predicts. The correct statement is that Table 9 carries no information about these models in either direction, and that the apparatus which produced it should not have been asked the question.

The question it should have been asked is whether the forecasts carry predictive content at all, and Table 10 answers it with the statistic that is correctly sized here. No model reaches significance: the Clark–West statistics run from -0.2286 for `rolling_mean` to 0.6214 for `ridge`, with one-sided p-values between 0.2672 and 0.5904, and Holm rejects nothing. The ordering is also worth noting, because it inverts the squared-error ordering: `ridge` and `previous_return`, two of the four models the squared-error test convicted, are the two that come closest to showing predictive content. That is what Equation 12 predicts—a forecast with more variance is punished harder regardless of its covariance—and it is the clearest illustration in this paper of why the two tests must be read together.

The right panel of Figure 8 makes the dependence problem concrete. At row level `logistic` reaches $p = 0.0015$; at session level it is 0.0541 and does not survive correction. Nothing about the data changed, only the claim about how much of it is independent. Table 2 says which of the two is right: at four names correlated at 0.5697 the row-level test rejects a true null 22.84% of the time and the session-level test 5.29%, so the median factor of 52 separating the two columns is a measurement of an error and not a difference of opinion.

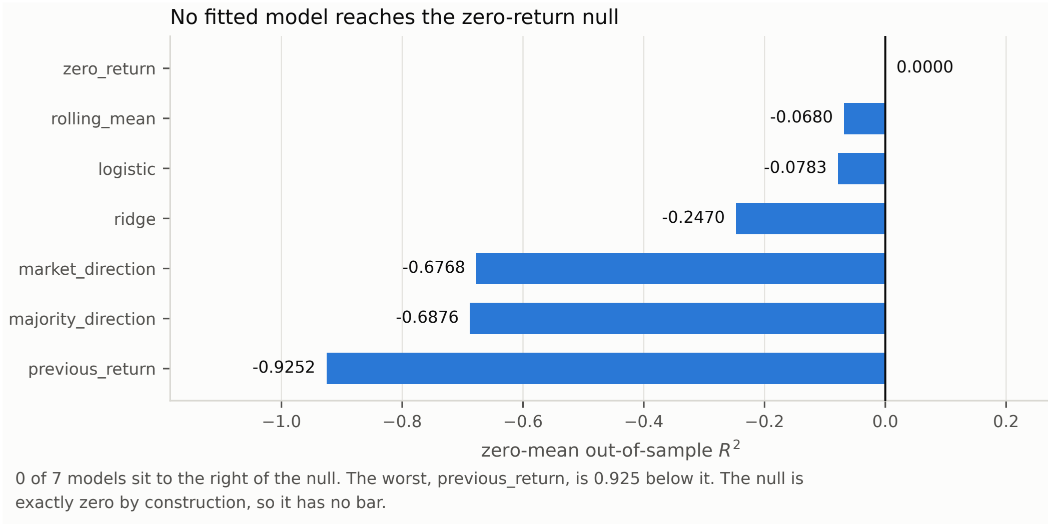


Figure 7: Zero-mean out-of-sample R^2 by model. The null is exactly zero by construction and therefore has no bar.

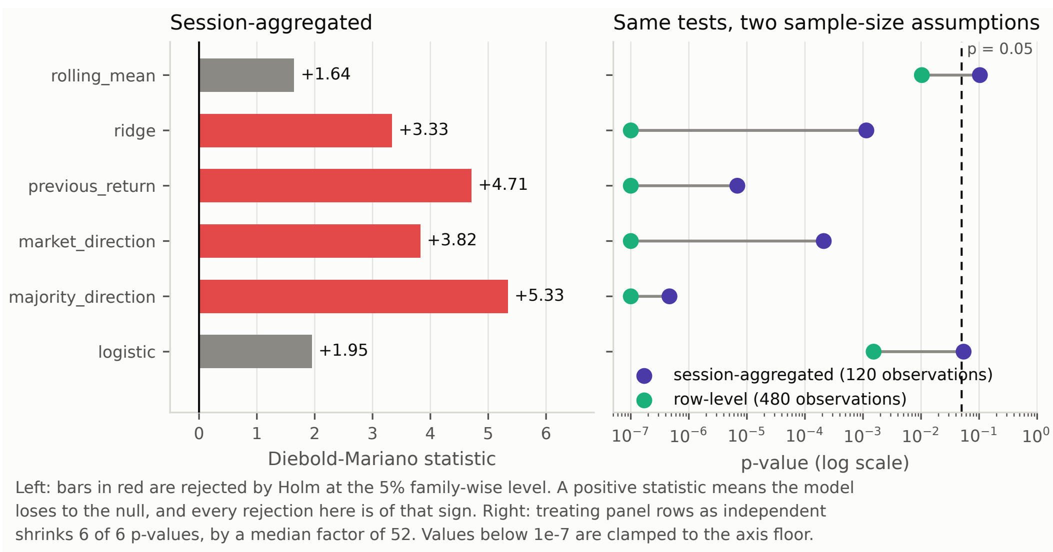


Figure 8: Diebold–Mariano statistics against the zero-return null, and the same six tests under two sample-size assumptions. Treating the 480 panel rows as independent shrinks every p-value, by a median factor of 52.

Directional accuracy deserves a word, because it is the metric this literature usually leads with. In this window 53.12% of realised targets are non-positive, so a constant "down" predictor attains 53.12% directional accuracy while carrying no information whatever. That is exactly the figure `zero_return` and `majority_direction` reach in Table 8, and it is above what `ridge`, `logistic` and `rolling_mean` manage. Directional accuracy without its base rate is not interpretable, which is why balanced accuracy is reported beside it and neither is used as a headline.

7.2 Nothing survives the search correction

Figure 9 is the data-snooping argument in one picture. The bootstrap null distribution is centred *above* zero because it is the distribution of a maximum over six candidates; that offset is precisely the correction, made visible. The observed maximum lies to the left of it and is itself negative, so

Table 11: Joint tests that no candidate beats the zero-return benchmark, over 10,000 stationary-bootstrap replications with mean block length 12.26. Hansen's three recentrings bracket the p-value.

Test	Statistic	p
White Reality Check	-0.000268	0.9970
Hansen SPA, lower	0.0000	0.6891
Hansen SPA, consistent	0.0000	0.6891
Hansen SPA, upper	0.0000	1.0000

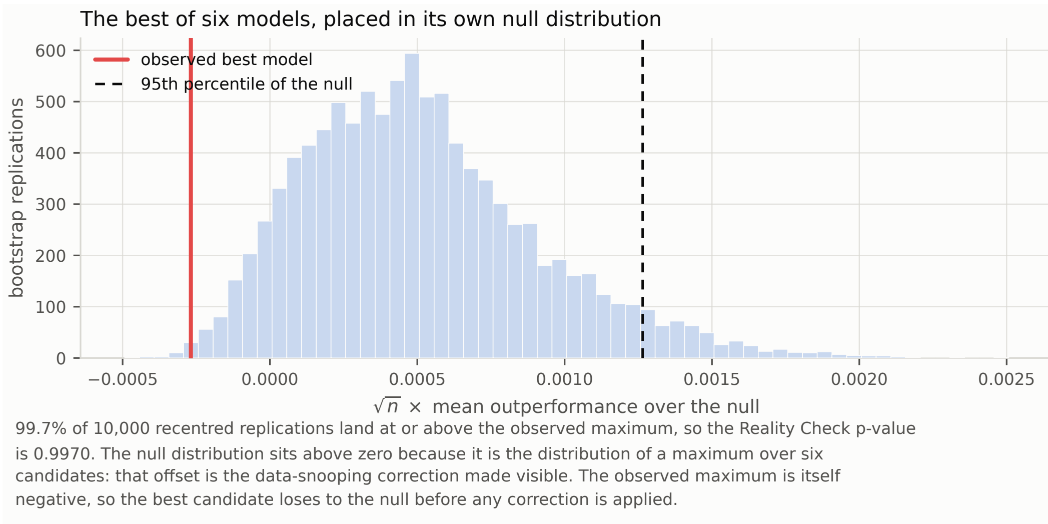


Figure 9: The best of six models placed inside its own bootstrap null distribution. The null is centred above zero because it is the distribution of a maximum over six candidates: that offset is the data-snooping correction made visible. The observed maximum is negative.

Table 12: Sharpe-ratio inference. The deflated ratio evaluates the probabilistic Sharpe ratio at the threshold the best of N skill-free configurations would be expected to reach.

Quantity	Value
Evaluated sessions	120
Per-session Sharpe	-0.0284
Annualised, square-root rule	-0.4513
Annualised, Lo autocorrelation-adjusted	-0.4973
Skewness	-0.0722
Kurtosis	7.2290
Probabilistic Sharpe ratio, threshold 0	0.3782
Configurations examined	72
Search threshold	0.1267
Deflated Sharpe ratio	0.0452

the best candidate loses to the null before any correction is applied. Table 11 gives the joint p-values: 0.9970 for the Reality Check and 0.6891 for Hansen's consistent recentring, bracketed by 0.6891 and 1.0000. The joint null is nowhere near rejection.

Table 12 carries the same message for the portfolio. Session returns have a kurtosis of 7.2290, so the normal-theory Sharpe interval understates tail risk; accounting for the higher moments, the probability that the true per-session Sharpe exceeds zero is 0.3782, already below even odds before any search correction. Deflating by the 72 configurations examined leaves a deflated Sharpe ratio of 0.0452.

Table 13: The configuration grid. Each trial is a complete re-run of the evaluation under one fold geometry and one portfolio breadth.

Quantity	Value
Configurations evaluated	72
Reported configuration rank by Sharpe	16
Median per-session Sharpe	-0.0682
Per-session Sharpe range	-0.1852 to 0.0356
Configurations with positive net return	5.56%
Best configuration net return	4.6259%
Null had the best R_0^2	72 of 72
Expected maximum Sharpe under no skill	0.1267

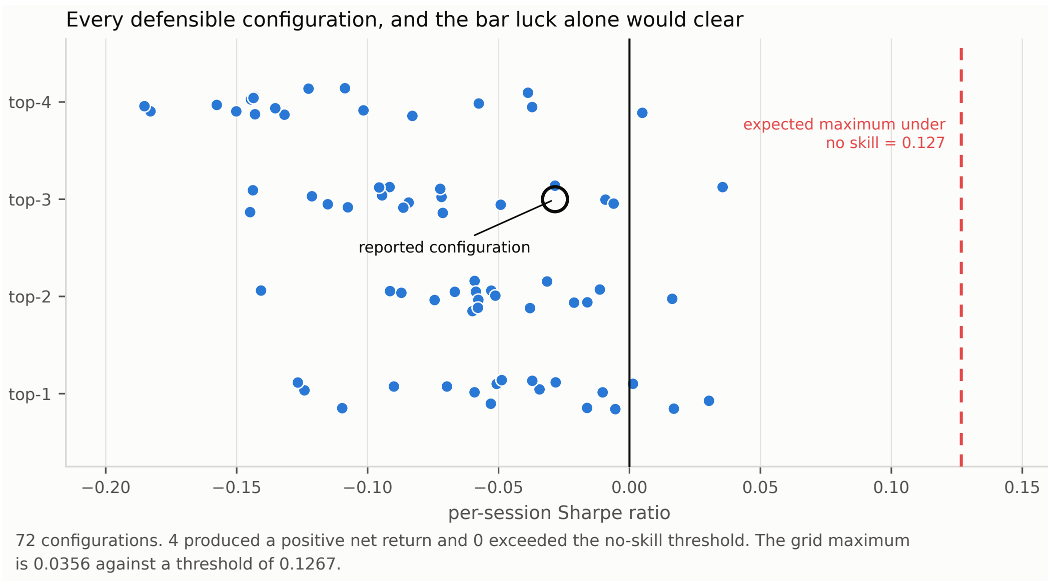


Figure 10: Every configuration in the grid against the Sharpe ratio that skill-free search alone would be expected to produce. The grid maximum does not reach the bar luck sets.

The grid itself varies the minimum training window, the validation width and step, the embargo and the portfolio breadth; each point is a complete re-run of the evaluation, not a re-weighting of a cached result (Table 13, Figure 10, and the full listing in Appendix E). Only 5.56% of those configurations produced a positive net return. The reported configuration ranks 16th of 72 by Sharpe ratio, so it is neither cherry-picked nor unusually unlucky, and the grid maximum of 0.0356 falls short of the 0.1267 that the best of 72 skill-free configurations would be expected to reach. The most compact statement of the result is that across all 72 configurations, the zero-return null attained the best out-of-sample R^2 in 72 of 72.

The deflated Sharpe ratio in the paragraph above rests on an assumption that Section 5.5 shows is both unnecessary and, as a test, invalid: its threshold is an expectation that half of all skill-free grids clear. Table 14 replaces it with a measurement. Recentring all 72 configurations to zero expected return and resampling them on a single stationary-bootstrap index draw preserves the co-movement the analytic correction has to guess at, and that co-movement is substantial: the mean pairwise correlation between configurations is 0.8395. The best of the grid, `train48_val20_step20_emb2_k2`, has a per-session Sharpe ratio of 0.0247 on the 87 sessions every configuration evaluated, against an expected maximum of 0.1049 and a 95th percentile of 0.2925 under the joint null. The exact bootstrap p-value is 0.7715.

The same bootstrap measures how much independent searching the grid represents, without the moment-matching that Proposition 5 requires: it behaves like 3.77 independent trials. The closed-

Table 14: The best of the configuration grid against a stationary bootstrap that recentres every configuration to zero expected return and resamples all of them on a single index draw. Unlike the deflated Sharpe ratio this assumes nothing about how the trials disperse, and it returns a critical value and a p-value rather than an expectation.

Quantity	Value
Configurations resampled jointly	72
Sessions common to every configuration	87
Mean pairwise correlation between configurations	0.8395
Stationary bootstrap block length	5.02
Best configuration	`train48_val20_step20_emb2_k2`
Its per-session Sharpe ratio	0.0247
Expected maximum under the joint null	0.1049
95th percentile under the joint null	0.2925
Exact bootstrap p-value	0.7715
Independent-equivalent trials	3.77

Table 15: The executed long-only top-k portfolio, computed entirely from the persisted event ledgers. Sessions holding risk is reported because annualised figures assume continuous deployment.

Measure	Value
Gross return	7.7694%
Net return	-4.7687%
Annualised return	-9.7521%
Annualised volatility	18.8297%
Sharpe	-0.4513
Maximum drawdown	-14.4181%
Turnover	123.68×
Round trips	143
Sessions holding risk	62 of 120
Equal-weight universe	-5.8309%
Modelled cost	TRY 12,403.77

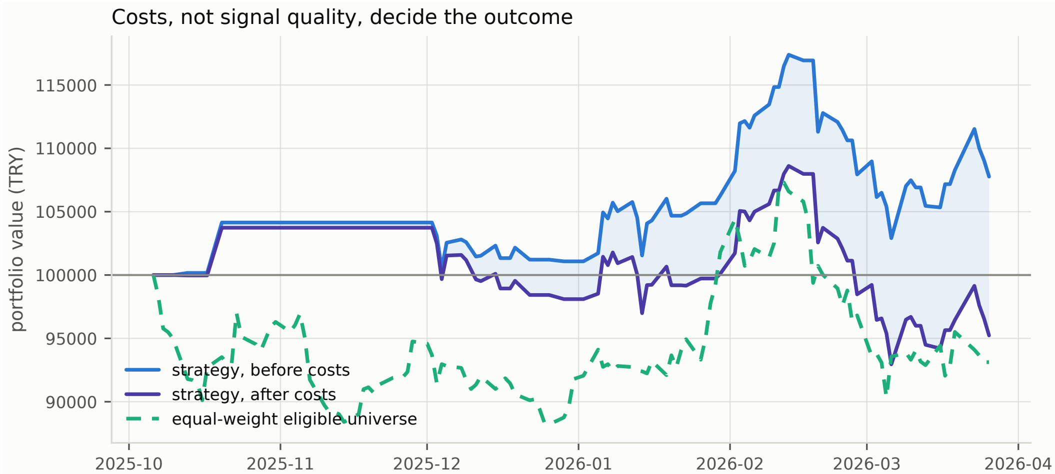
Table 16: Cost sensitivity. The same trading decisions are reused across the three cases, so only the bill changes; net return cannot improve as costs rise.

Case	Gross	Net	Total cost	Round trips
0.0x	7.7671%	7.7671%	TRY 0.00	143
1.0x	7.7694%	-4.7687%	TRY 12,403.77	143
2.0x	7.7188%	-15.8890%	TRY 23,338.11	143

form route of Section 6.4 gives 7.01. Both are far below the nominal 72 and the conclusion does not turn on which is preferred, but the gap is itself worth recording, since the closed form errs towards claiming more independence than the grid has, and therefore towards a harsher correction than necessary. The direction is conservative, which is the direction one wants an error in a search correction to run.

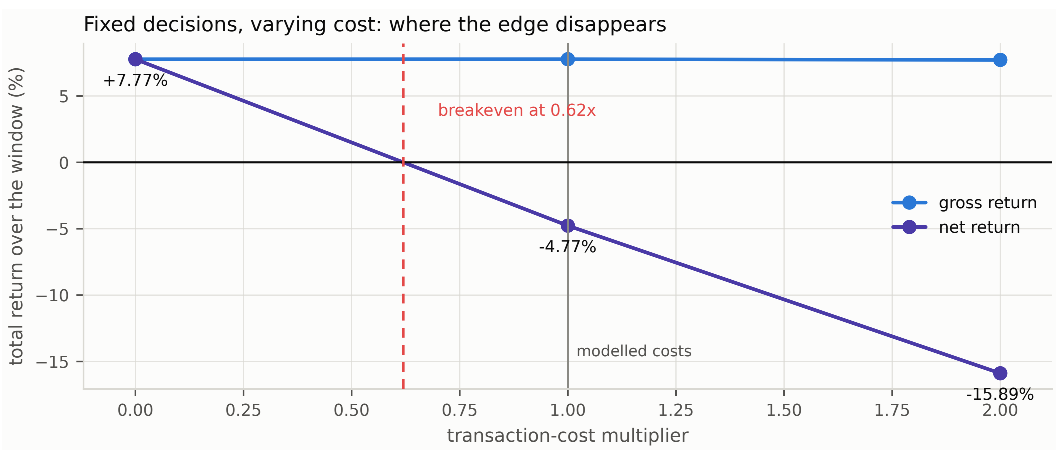
7.3 Costs, not signal, decide the outcome

The strategy earns a gross return of 7.77% and loses 4.77% of its capital after costs, on $123.7\times$ turnover and TRY $12{,}404$ of modelled cost (Table 15). The three cost cases hold the trading *decisions* fixed—the same 143 round trips in each—but re-simulate the capital path, so gross return is not exactly invariant across them: it moves by 5×10^{-4}, because position sizing responds to the cash the previous session's costs consumed. A strictly frozen ledger, in which only the debited bill changes, is the cleaner counterfactual and is not what is computed here. Table 16 should be read accordingly: net return falls monotonically, and interpolating linearly between the zero-cost and



Gross +7.77%, net -4.77%: TRY 12,404 of modelled costs on 123.7x turnover. The flat stretches are sessions with no position: expected net return was non-positive for every eligible name, so the strategy holds risk on only 62 of 120 sessions (52%). Any figure annualised as if the capital were continuously deployed therefore overstates the deployment.

Figure 11: Gross and net equity against the equal-weight eligible universe. The flat stretches are sessions with no position: the strategy holds risk on 62 of 120 sessions, so annualised figures overstate the deployment.



Decisions are held fixed across the three cases, so only the bill moves; net return is monotonically decreasing in the cost multiplier. Breakeven sits at 0.62x the modelled cost, so the gross edge is exhausted below the costs an actual desk would pay.

Figure 12: Fixed trading decisions under three cost multipliers. Breakeven sits at $0.62\times$ the modelled schedule.

modelled-cost cases puts breakeven near $0.62\times$ the modelled schedule, which is an approximation through a nonlinear simulation rather than an exact root. The schedule itself is a declared scenario, not a desk quote calibrated to BIST spreads and depth. This reproduces Novy-Marx and Velikov [35] at small scale, with one caveat the phrase usually elides: a positive realised gross return over 120 sessions is a sample outcome, not an established population edge, and the forecast behind it is not distinguishable from the null.

Figure 11 shows something the return statistics hide. The flat stretches are sessions in which no eligible name had a positive expected net return, so the strategy held nothing; it carries risk on 62 of 120 sessions. Any figure annualised as though the capital were continuously deployed, including the annualised return and Sharpe ratio in Table 15, therefore overstates the deployment and is reported only because it is the convention.

Table 17: The 95% bootstrap interval for annualised return, repeated across every candidate block length.

Block length	Lower	Upper	Contains zero
1	-47.01%	53.29%	yes
2	-46.03%	50.55%	yes
3	-45.41%	48.56%	yes
5	-45.60%	48.28%	yes
8	-47.09%	46.13%	yes
13	-49.26%	48.93%	yes
21	-47.01%	43.24%	yes

7.4 The conclusion does not depend on the arbitrary choices

A bootstrap block length is an arbitrary choice, so all seven candidates are reported rather than one (Table 17, Figure 13). Every interval contains zero and their widths vary by about ten percentage points out of ninety, so the interval is a statement about the data rather than about the block length. The automatic Politis–White selection lands at 2.12 sessions for the portfolio return series and at 12.26 for the loss differentials, which is why the two bootstraps in this paper carry different block lengths.

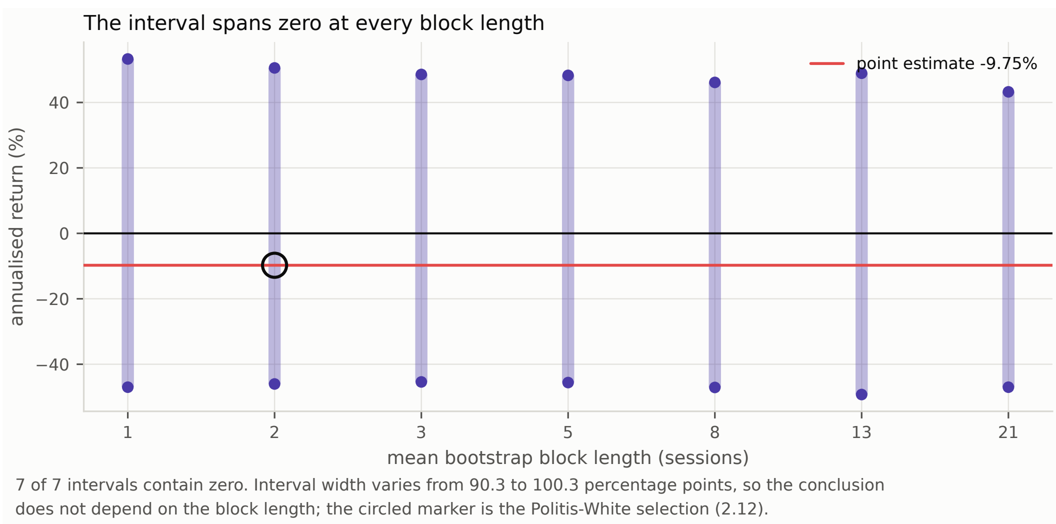


Figure 13: The 95% bootstrap interval for annualised return across seven block lengths. Every interval contains zero.

7.5 The design was never powered for a plausible effect

Table 18 inverts the tests that were actually run. At 120 sessions, the best-powered candidate could have been separated from the null only by an out-of-sample R_0^2 above 0.1132. That is roughly eleven times the upper end of what Campbell and Thompson [27] and Gu et al. [28] treat as economically meaningful, and about a hundred times the values those studies typically report. Detecting $R_0^2 = 0.01$ at the same size and power would need $15{,}126$ sessions of this panel: about sixty years of trading.

The portfolio side is worse. Because the search threshold does not shrink with the record length, establishing skill at the conventional deflated-Sharpe bar of 0.95 would have required a per-session Sharpe ratio of 0.2884, an annualised 4.5785. No equity strategy operates there. The correct reading of the deflated Sharpe ratio of 0.0452 is therefore not “the strategy has no skill” but “a 120-session record searched over 72 configurations cannot demonstrate skill at any plausible level, and this one does not come close either”.

Table 18: Inverting the accepted tests: the smallest effects this design could have separated from the null at conventional size and power.

Quantity	Value
Best-powered candidate	logistic
Standard error of its session-mean loss differential	1.444e-05
Test size α	0.05
Target power	0.80
Smallest detectable loss differential	4.079e-05
Mean squared error of the null	3.604e-04
Smallest detectable out-of-sample R_0^2	0.1132
Reference effect assumed plausible	0.0100
Sessions required for that effect	15,126
Per-session Sharpe required for a deflated ratio of 0.95	0.2884
Its annualised equivalent	4.5785

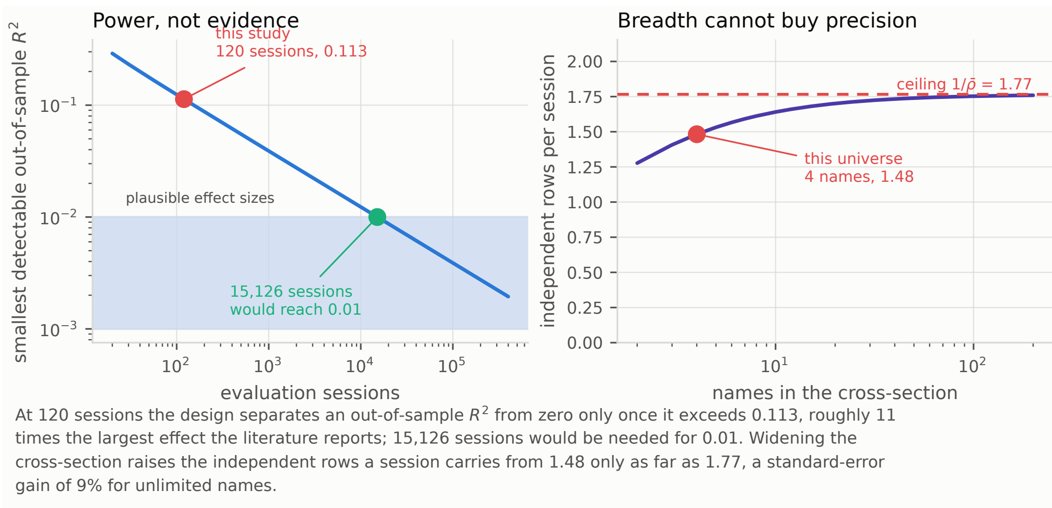


Figure 14: Left: the smallest out-of-sample R^2 separable from zero at 5% size and 80% power, as a function of the number of evaluation sessions, with this study marked and the band of effects the literature reports shaded. Right: independent rows per session against the number of names, with the $1/\bar{\rho}$ ceiling of Proposition 2.

The right panel of Figure 14 closes the obvious escape route. Proposition 2 caps the independent rows a session can carry at $1/\bar{\rho} = 1.7554$; this four-name panel already reaches 1.4765 of that. An unlimited universe at the same correlation would improve standard errors by 9%—which moves the detectable R_0^2 from 0.113 to about 0.095, not to 0.01. Breadth is not the missing ingredient. Sessions are.

7.6 The cost schedule rules the design out before any model is chosen

Evaluating Equation 16 on the accepted design gives the sharpest result in this paper (Table 19). At a round-trip cost of 20.2 basis points against a target standard deviation of 1.90%, holding the top 3 of 4 names—a selection score of only $\lambda(4, 3) = 0.3431$, since holding three quarters of the universe barely selects at all—requires a cross-sectional information coefficient of 0.3098. The ridge model achieved 0.0092 per session, with a standard error of 0.0549: indistinguishable from zero. The design demanded roughly thirty times the skill it got.

The correlation pooled over stock-sessions is 0.0387, four times larger, and it is the number a study would quote without distinguishing the two. It is not the quantity ranking uses: it also absorbs the

Table 19: Proposition 3 evaluated on the accepted design. The requirement depends on the cost schedule, the volatility of the target and the breadth of the rule, and on nothing the forecaster does. Selection happens within a session, so the per-session cross-sectional IC is what instantiates ρ; the pooled correlation is larger because it also absorbs the common time-series component that ranking cannot use. The final row is the fixed-count breadth at which the bound would be met: λ is unbounded in N, so this is a statement about attainable universes, not an impossibility result.

Quantity	Value
Round-trip cost on notional	20.20 bp
Standard deviation of the executable target	1.90%
Names ranked	4
Names held	3
Selection score $\lambda(N, k)$	0.3431
Information coefficient required	0.3098
Per-session cross-sectional IC achieved	0.0092 ($\pm$ 0.0549)
Shortfall factor	33.6$\times$
Pooled IC over stock-sessions, for contrast	0.0387
Universe clearing the bound while holding one name	$> 10^6$

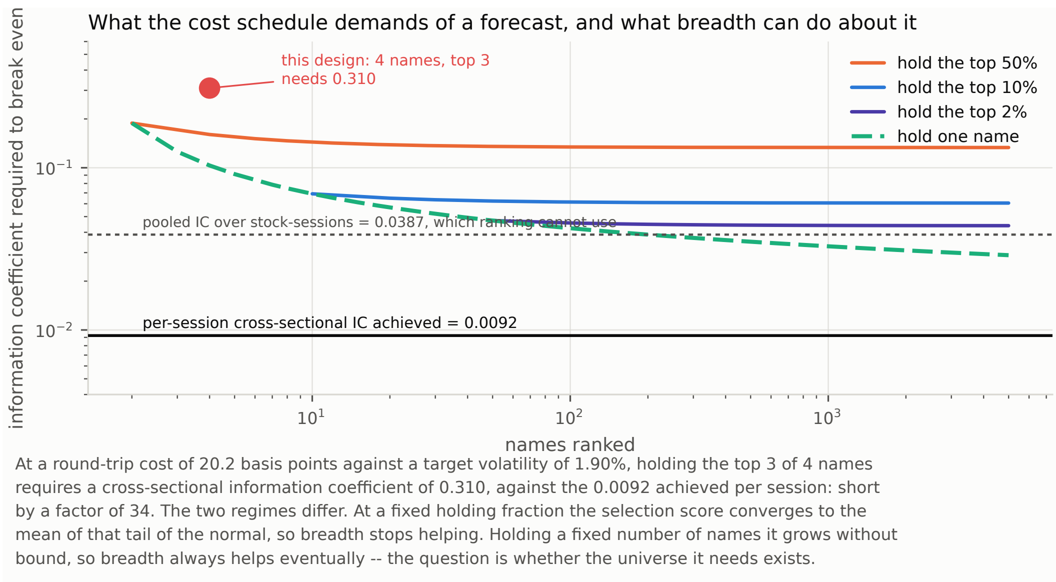


Figure 15: Proposition 3 evaluated across universe widths and holding fractions at this run's cost schedule. The information coefficient the design demands is compared against the one the portfolio model achieved.

common time-series component, so a forecast that merely tracks the market direction earns it while ordering nothing within a session.

Figure 15 shows what breadth can and cannot do, and the two regimes of Proposition 4 answer differently. At a fixed holding fraction the requirement converges: holding the top 2% of 500 names still needs about 0.043. At a fixed holding count it does not converge, so a sufficiently wide ranking always clears the bound—at the pooled correlation, a few hundred names ranked with one held would already suffice. Measured at the per-session coefficient the universe required exceeds 10^6 names. That is the defensible form of the claim: not that breadth cannot help, but that no attainable market is wide enough at this level of skill, and that the single-name rule which comes closest is exactly the one capacity and market impact exclude.

This is worth separating from the statistical result, because the two are independent. The tests in Section 7.2 say the forecast is not distinguishable from zero. Proposition 3 says that even a forecast that *was* distinguishable from zero—at, say, an information coefficient of 0.05, which would be a

Table 20: The False Strategy threshold under both readings of the grid. The best configuration falls below either, so the conclusion does not turn on how the trials are counted.

Quantity	Value
Configurations searched	72
Realised variance of trial Sharpe ratios	0.002757
Variance implied by independent trials	0.008337
Effective independent trials	7.01
Threshold at the realised dispersion	0.1267
Threshold if the trials were independent	0.2203
Best configuration in the grid	0.0356

respectable result in this literature—would still have lost money in this design. A study that reported such a forecast as a success, without evaluating the bound, would be reporting a statistically real and economically worthless effect.

7.7 Every reading of the search correction agrees

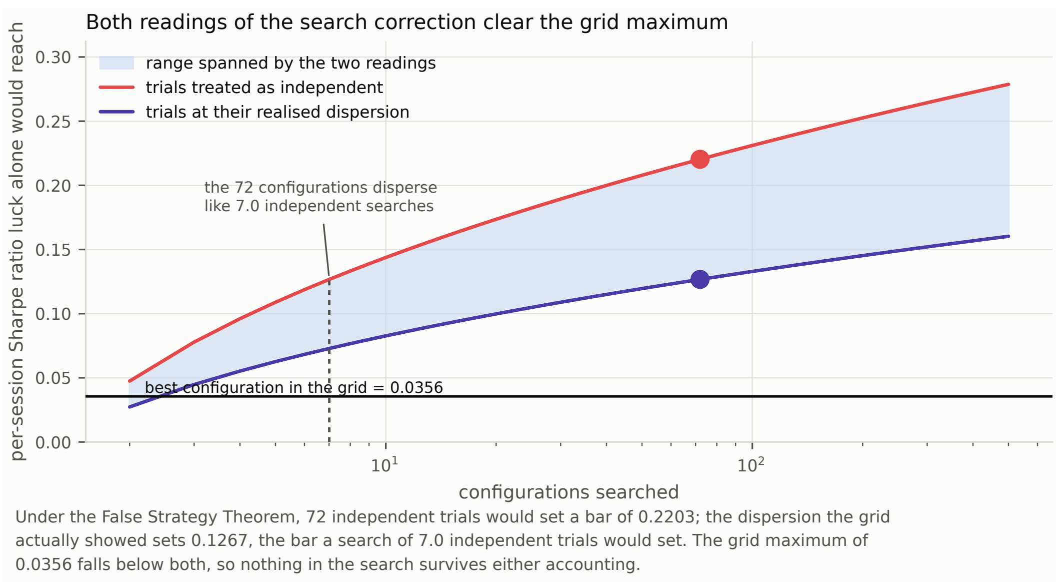


Figure 16: The False Strategy threshold as a function of the number of configurations searched, under independent trials and at the dispersion the grid actually showed. The vertical marker is N_{eff} from Proposition 5.

The 72 configurations disperse considerably less than 72 independent trials would: their realised Sharpe variance is 0.002757 against the 0.008337 a single 120-session estimate carries. Solving Equation 17 puts $N_{\text{eff}} = 7.01$ (Table 20). The grid, in other words, asked about seven genuinely different things while appearing to ask seventy-two. That is a useful diagnostic in its own right: a search whose effective count is far below its nominal count is exploring one design repeatedly rather than exploring a design space.

For the conclusion it matters less than one might expect, which is the point of reporting every reading. Treating the trials as independent sets the bar at 0.2203; using their realised dispersion sets it at 0.1267. The grid maximum is 0.0356, below both by a wide margin (Figure 16), and the joint bootstrap of Table 14—which sets no bar at all but computes a p-value directly—returns 0.7715. Whatever one believes about how to count a correlated search, nothing in this grid survives it.

The three routes disagree about the effective trial count, and the disagreement is instructive rather than embarrassing. Moment-matching against Lo's sampling variance gives 7.01; the joint bootstrap gives 3.77. The first substitutes the realised dispersion of trial Sharpe ratios for a sampling variance, which conflates true heterogeneity between configurations with estimation noise; the second measures the distribution of the maximum directly and needs no such substitution. Where they differ we report both and prefer the second, and Section 5.5 is the reason: the bootstrap recovers a nominal 72 when the trials really are independent, which is the only check that distinguishes a diagnostic from a number.

8 Discussion

8.1 What this result is, and is not

On this universe, over this window, at this horizon, scale-normalised price and volume features carry no out-of-sample information about next-session open-to-close returns that survives an honest accounting of sample size, dependence, nesting and search. The family is far from rejection jointly, no forecast shows predictive content under the one test that is correctly sized for the comparison, and the best of 72 configurations sits at a joint p-value of 0.7715.

It does not follow that Borsa Istanbul is efficient, that no short-horizon signal exists, or that machine learning cannot forecast equity returns. It does not even follow that this particular signal is absent. Section 7.5 makes the honest version of the claim available: an effect large enough to be *usable* at this horizon on this universe would have shown up, and did not; an effect of the size the literature actually reports would not have shown up either way, and the experiment is silent about it.

Two details deserve emphasis, and both are corrections to what a reader would otherwise reasonably conclude. First, a model with no signal does not merely fail to help; it adds estimation variance to the forecast, which is a pure cost in a squared-error comparison against a zero forecast. It is tempting to read several fitted models falling below zero as a consistency check confirming they are worthless. It is not a check of anything: Section 5.3 shows the same pattern arises with probability 0.9219 when the models are in fact exactly at the null, so the observation has no diagnostic value and must be replaced by the Clark–West test rather than interpreted. Second, the False Strategy threshold is an *expectation*, not a critical value: a genuinely skill-free grid exceeds it in 45.7% to 50.0% of replications regardless of how dependent the trials are. It is a diagnostic, and the test is either the deflated Sharpe ratio, which converts it into a probability under an independence assumption, or the joint bootstrap, which does not need one.

8.2 What follows for the design of such studies

The checks in Sections 5 and 6 are cheap to compute and, we would argue, belong in the design phase rather than the post-mortem. Most of them can be evaluated before any data is modelled.

Compute the size of your test before trusting its p-values. Equation 11 needs two numbers a panel study already has: the width of the cross-section and the average correlation within it. A study pooling twenty names correlated at 0.4 is running a 57% test and calling it a 5% test. This is the cheapest check in the paper and the one with the largest consequences.

Aggregate to the unit at which observations are independent, and say so. The gap between the row-level and session-level p-values here is a median factor of 52, and Section 5.2 establishes which side of that gap is correct. A panel study that does not state its unit of inference has not reported its result.

Do not compare a fitted model against a zero or mean benchmark under squared-error loss. They are nested, and the comparison is decided by the fitted model's own variance. Use Clark–West, or change the null. If a study reports that its models are significantly worse than a naive benchmark, that finding is very likely an artefact and should not be interpreted at all.

Compute the feasibility bound before choosing a model family. Equation 16 needs only the cost schedule, the target volatility and the intended breadth. It is the difference between "our model was not good enough" and "no model would have been", and in this study it is decisive. Reporting it also disciplines the reader: a paper that clears the statistical bar but not the feasibility bound has found something true and useless, and should say so.

Compute the detectable effect before running the experiment. Equation 13 needs only the target's variance and the intended sample size. A design whose detectable effect exceeds the plausible effect by an order of magnitude will produce an uninformative answer whichever way it comes out, and should be lengthened or abandoned rather than run.

Report the effective size of the search, not only its nominal size, and resample the grid rather than deflating it. N_{eff} between 3.77 and 7.01 against a nominal 72 says the grid was far narrower than it looked. The opposite case—a search whose configurations are genuinely independent—deserves a correspondingly harsher correction, and neither is visible from the trial count alone. A joint bootstrap costs one index draw per replication and returns a p-value instead of a comparison against an expectation.

8.3 Verifying the evaluator, not only the model

An evaluation apparatus is software, and software with no adversarial test is software of unknown quality. Twenty real defects in the inference and reporting code—an autocovariance normalised by $n - k$ instead of n, a Holm correction without its running maximum, a Diebold–Mariano test aggregating rows instead of sessions, a deflated Sharpe ratio that ignores the trial count, a Clark–West adjustment missing its factor of two, a joint bootstrap that resamples each configuration separately, and fourteen others—are reintroduced one at a time by a harness that requires a named test to fail for each and to pass again once the edit is reverted (Appendix F). Four of the guarding tests turned out to be decorative and were repaired as a result: three when the catalogue was first written, and one when the joint search was added, where a skill-free grid could not detect a missing recentring because its means were already near zero.

We think this is the right standard for statistical code that produces published numbers, and it is cheaper than it sounds: the catalogue is a list of edits and test identifiers, and the harness is under a hundred lines. A negative result in particular has no external check—nobody notices when a broken test fails to find an effect that is not there—so the apparatus has to be checked directly.

Mutation testing and simulation calibration answer different questions and a serious pipeline needs both. Mutation testing asks whether the code implements the estimator its author intended; simulation calibration asks whether that estimator answers the question its user believes it answers. Everything Section 5 found wrong was correctly implemented. The Diebold–Mariano test computes the Diebold–Mariano statistic exactly as specified, on rows that are not independent, for models that are nested. No amount of testing against the specification would have found it, because the specification was not the problem.

9 Limitations

Four stocks and 120 evaluated sessions are too few for a strong economic conclusion, and Section 7.5 quantifies exactly how few. The universe is a fixed prototype rather than point-in-time index membership. The committed input contains no BIST index series, so only cash and an equal-weight eligible-universe benchmark are reported and no index-relative claim is made. Sector metadata is unavailable and is reported as unavailable rather than filled with a placeholder. The strategy holds risk on 62 of 120 sessions, so annualised figures assume a deployment that did not occur. The search grid is itself a choice: a different grid would give a different deflation threshold, and the grid is declared in the run configuration so the choice is auditable. Corporate-action transition and fail-closed policies are exercised by synthetic tests; the empirical snapshot contains cash dividends only.

The detectability results carry their own assumptions. Proposition 2 uses an equicorrelated approximation, which is a simplification of a correlation matrix whose realised pairwise values here range from 0.46 to 0.83; the ceiling it gives is an average-case statement, and it holds the average correlation fixed as the universe expands, which a real expansion would not. It is instantiated on the dependence of the loss differential rather than of the raw target, since that is what enters the standard error of the test; the two are close here but they are not the same object. Proposition 3 assumes joint normality, an unbiased forecast and an iid cross-section, and ignores estimation error in the ranking—all of which make the bound optimistic, so a design that fails it fails a fortiori, while a design that clears it has met a necessary condition only.

The closed-form power calculation of Equation 13 is the weakest of the three. It treats the loss-differential standard error as scaling exactly as $n^{-1/2}$, which is exact only if the dependence and the higher moments of the differential are stable across window lengths—an assumption a sixty-year extrapolation strains badly. It inverts the test with central t quantiles rather than the noncentral distribution, and it instantiates the calculation at the standard error of the candidate selected *after* seeing the run. The resulting session counts are order-of-magnitude statements about the design rather than sample-size requirements. Table 7 supersedes it where the two overlap, by raising a known population effect until the measured rejection rate reaches 80%; it reports a larger minimum detectable effect than the closed form, which means the closed form is optimistic and the design was even less well powered than Section 7.5 states.

The calibration study has limits of its own, set out in Section 5 and repeated here because they bound every claim in this paper. Its panels are equicorrelated rather than factor-structured, its forecast families share one predictable component and differ only in noise, and its search experiment imposes exchangeable dependence between trials. Each simplification omits heterogeneity that a real design has, and in each case the omission works in the direction of understating the measured error rate, so the numbers should be read as lower bounds. The study also calibrates the estimators against panels matched to *this* panel; a study whose cross-sectional correlation or forecast variance ratio differs must re-run the calibration at its own design point rather than citing these numbers, which is why the generator and the study driver are part of the released code.

Finally, the repository implements Kalman filters, Ornstein–Uhlenbeck mean reversion, GARCH, HMM regimes, wavelets, cointegration, Kelly sizing, gradient boosting, stacking, calibration and neural sequence models, none of which is part of the accepted experiment; their presence is not evidence of predictive value, and a machine-readable component status table records for each one whether it is integrated, evaluated, and backed by run evidence.

10 Conclusion

The standard apparatus for comparing forecasts on an equity panel is miscalibrated in three places, and we measured how badly. Applied to panel rows, the equal-accuracy test rejects a true null 22.84% of the time on four correlated names and 79.13% on a hundred, at a nominal 5%; the inflation has the closed form $2\Phi(-z_{1-\alpha/2}/\sqrt{1+(k-1)\bar{\rho}})$, which the simulation reproduces to within 0.0064 everywhere, and Holm's family-wise error rate follows it to 58.75%. Compared against a zero benchmark under squared-error loss, a forecast that is pure estimation noise is declared significantly worse than the benchmark in 92.19% of replications, rising to certainty as the record lengthens. The False Strategy Theorem's threshold is cleared by roughly half of all skill-free grids at every level of dependence between trials, because it is an expectation. Each failure has a replacement that holds its nominal level under heavy tails, clustered volatility, regime switches and record lengths from 60 to 500 sessions: session aggregation, the Clark–West statistic, and a synchronised bootstrap of the whole grid.

Applied to a leakage-controlled Borsa Istanbul panel with an executable target, the corrected apparatus returns a negative result, and a different one from what the uncorrected apparatus reports. No forecast shows predictive content; the joint data-snooping null is not rejected; the best of 72 configurations—which behave like fewer than four independent trials—has an exact joint p-value of 0.7715. What does *not* survive is the claim that four of the six models are significantly worse than the null. That is the nesting artefact, and a study reporting it as a finding would be reporting the output of its own estimation variance.

Inverting the corrected tests turns the negative result from a verdict about the market into a verdict about the experiment. The design could only have resolved an out-of-sample R^2 above 0.113 in closed form and 0.201 by simulation, against the 0.01 this literature treats as economically meaningful. Widening the cross-section substitutes poorly, because the loss differentials of correlated names carry at most $1/\bar{\rho}$ independent observations per session. And the cost schedule demanded an information coefficient of 0.310 from a design that delivered 0.009 per session—a bound no choice of model family could have met, and one computable before any model was fitted.

The useful output is therefore the apparatus and its self-assessment: a target the backtest can trade, an evaluation whose unit of inference matches the data's dependence structure, tests whose measured size is reported alongside their nominal size, a search that is resampled rather than assumed, an explicit statement of what the design could and could not have found, and a run that replays byte-for-byte. Two next steps follow, and they are not the same step. For this market, a longer point-in-time dated universe and a cheaper execution assumption, chosen so that the feasibility bound is cleared before the first model is fitted—not another model family. For the literature, a calibration run at each study's own design point, which costs a few minutes of computation and is the difference between a p-value and a number that resembles one.

Reproducibility

Every table and figure in this manuscript is generated by a tested module from one of two immutable artifacts—the run bundle for the empirical sections, the calibration study for Section 5—and regenerated by `make paper`; the manuscript cannot state a table entry the pipeline does not produce. The run replays byte-for-byte from bundled inputs with `make reproduce-committed`. The calibration study is rebuilt by `bist-predict calibrate`, takes about four minutes, and carries a content hash over its own numbers that excludes the elapsed time, so the artifact is identified by what it measured rather than by the machine that measured it. `make verify-claims` checks the prose figures in this manuscript and in the repository's README against those artifacts, and `make mutation-check` reintroduces twenty real defects into the source and requires the guarding test to fail for each. Code and artifacts: https://github.com/ITheClixs/Prediction-System-for-Istanbul-Stock-Exchange.

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A Proofs

Proof of Proposition 1. Let $d_{i,t}$ have common variance σ^2 and equicorrelation $\bar{\rho}$ within a session, and be independent across the T sessions. The pooled mean $\bar{d}$ over kT observations has true variance

$$\operatorname{Var}(\bar{d}) = \frac{\sigma^2}{kT}\left(1 + (k-1)\bar{\rho}\right), \tag{18}$$

by the computation in the proof of Proposition 2. A test that divides by $\sqrt{\hat{\sigma}^2/(kT)}$ uses a denominator that converges to $\sigma/\sqrt{kT}$, so the statistic it forms is $\sqrt{1 + (k-1)\bar{\rho}}$ times a standard normal variate in the limit. Writing $\kappa = \sqrt{1 + (k-1)\bar{\rho}}$, its two-sided rejection probability is

$$\Pr\left[|\kappa Z| > z_{1-\alpha/2}\right] = 2\,\Phi\!\left(-\frac{z_{1-\alpha/2}}{\kappa}\right), \tag{19}$$

which is Equation 11. Since Φ is strictly increasing and $-z_{1-\alpha/2}/\kappa$ is strictly increasing in κ, and κ is strictly increasing in k for $\bar{\rho} > 0$ and in $\bar{\rho}$ for $k > 1$, the size is strictly increasing in both. Finally $\kappa \to \infty$ as $k \to \infty$ whenever $\bar{\rho} > 0$, so the argument of Φ tends to 0 and the size tends to $2\Phi(0) = 1$. □

Remark 3. The proposition is stated for the dependence of the *loss differential*, which is what enters the standard error of the test, rather than of the raw target. The two are close on the committed panel—0.5662 against 0.5697—but they are different objects, and only one of them governs the size.

Proof of Proposition 2. Let $x_1, \ldots, x_k$ be the session's targets, each of variance σ^2, with $\operatorname{corr}(x_i, x_j) = \bar{\rho}$ for $i \neq j$. The variance of their mean is

$$\operatorname{Var}\left(\frac{1}{k}\sum_i x_i\right) = \frac{\sigma^2}{k^2}\left(k + k(k-1)\bar{\rho}\right) = \frac{\sigma^2}{k}\left(1 + (k-1)\bar{\rho}\right), \tag{20}$$

so the session mean is as informative as $m(k) = k/(1 + (k-1)\bar{\rho})$ independent observations. Differentiating,

$$m'(k) = \frac{1-\bar{\rho}}{(1+(k-1)\bar{\rho})^2} > 0 \quad \text{for } \bar{\rho} < 1, \tag{21}$$

so m is strictly increasing, and dividing numerator and denominator by k gives $m(k) \to 1/\bar{\rho}$ as $k \to \infty$. Since m is increasing and convergent, the limit is a supremum. $\square$

Proof of Proposition 3. Write $\hat{r}$ in standardised form $z = (\hat{r} - \mu_{\hat{r}})/\sigma_{\hat{r}}$. For a bivariate normal pair with correlation ρ,

$$\mathbb{E}[r \mid z] = \rho\, \sigma\, z. \tag{22}$$

Selection holds the k names with the largest $\hat{r}$, which is a function of the standardised scores alone; conditional on the ranking, the selected names have standardised scores distributed as the top k order statistics of N standard normal draws. By the tower property the expected realised return of an equally weighted holding is

$$\mathbb{E}\left[\frac{1}{k}\sum_{i=N-k+1}^{N} r_{(i)}\right] = \rho\,\sigma\,\frac{1}{k}\sum_{i=N-k+1}^{N} \mathbb{E}\big[Z_{(i:N)}\big] = \rho\,\sigma\,\lambda(N,k). \tag{23}$$

Charging the round-trip cost c on the notional of each position, the expected net return per name is $\rho\,\sigma\,\lambda(N,k) - c$, and requiring it to be positive gives Equation 16. The second form of λ in Equation 15 follows from exchangeability: a given draw lies among the top k exactly when at most $k-1$ of the remaining $N-1$ draws exceed it, so

$$\begin{aligned}\sum_{i=N-k+1}^{N} \mathbb{E}\big[Z_{(i:N)}\big] &= N\,\mathbb{E}[Z \cdot \mathbf{1}\{Z \text{ in top } k\}] \\ &= N\int_0^1 \Phi^{-1}(u)\ \Pr[\mathrm{Bin}(N-1,\,1-u) \le k-1]\,\mathrm{d}u, \end{aligned} \tag{24}$$

and the binomial tail is the regularised incomplete beta function $I_u(N-k,\,k)$. $\square$

Proof of Proposition 4. For fixed q, the empirical distribution of the top $\lceil qN \rceil$ of N standard normal draws converges to the conditional law of Z given $Z > \Phi^{-1}(1-q)$, whose mean is $\varphi(\Phi^{-1}(1-q))/q$ by direct integration; λ is that mean, so it converges to the same finite limit. For fixed k, $\lambda(N,k) \ge \mathbb{E}[Z_{(N-k+1:N)}]$, and the smallest retained order statistic already diverges: for $k = 1$ the classical extreme-value result for the Gaussian maximum gives $\mathbb{E}[Z_{(N:N)}] = \sqrt{2\log N}\,(1+o(1))$. Hence $\lambda(N,k) \to \infty$ and $c/(\sigma\lambda(N,k)) \to 0$, so for any $\rho > 0$ there is an N satisfying Equation 16. $\square$

Proof of Proposition 5. Φ^{-1} is strictly increasing, and $1 - 1/N$ and $1 - 1/(Ne)$ are both strictly increasing in N on $(1,\infty)$; q is a convex combination of the two with positive weights $1-\gamma$ and γ, hence strictly increasing and continuous there, with $q(N) \to -\infty$ as $N \downarrow 1$ and $q(N) \to \infty$ as $N \to \infty$. The right-hand side of Equation 17 is therefore a continuous strictly increasing bijection from $(1,\infty)$ onto $\mathbb{R}$, so for any fixed left-hand side the equation has exactly one solution. Bisection converges to it. $\square$

Remark 4. Proposition 5 identifies N_{eff} by matching expected maxima, not by matching the full distribution of the maximum. Two searches with the same expected maximum can differ in the tail, so N_{eff} should be read as a summary of how much independent exploration the grid represents rather than as a claim that the grid is distributionally equivalent to N_{eff} independent trials.

B Feature manifest

Table 21: The accepted feature manifest, schema version `2.0.0`, digest `b0349ccf178ddaf1`. Every entry declares its formula, its lookback in sessions, and a normalisation policy of `none`: the accepted models consume scale-free quantities, so nominal price level cannot act as accidental ticker identification. A panel built against a different manifest digest is refused.

#	Feature	Formula	Lookback
1	`log_return_1d`	`log(split_close_t / split_close_t_minus_1)`	2
2	`log_return_5d`	`log(split_close_t / split_close_t_minus_5)`	6
3	`log_return_20d`	`log(split_close_t / split_close_t_minus_20)`	21
4	`close_over_sma20_minus_1`	`split_close / mean_20(split_close) - 1`	20
5	`sma20_over_sma100_minus_1`	`mean_20(split_close) / mean_100(split_close) - 1`	100
6	`atr14_over_close`	`atr_14(split_adjusted_ohlc) / split_close`	15
7	`vwap20_over_close_minus_1`	`vwap_20(split_adjusted_hlc, raw_volume) / split_close - 1`	20
8	`log_volume`	`log1p(raw_volume)`	1
9	`volume_zscore_20`	`zscore_20(raw_volume)`	20
10	`realized_volatility_20`	`std_20(log_return_1d) * sqrt(252)`	21
11	`intraday_range_over_close`	`(split_high - split_low) / split_close`	1
12	`overnight_gap`	`split_open / prior_split_close - 1`	2
13	`drawdown_20`	`split_close / max_20(split_close) - 1`	20
14	`cross_sectional_return_rank`	`midrank_percentile_by_date(log_return_20d)`	21
15	`market_relative_return_20d`	`log_return_20d - date_mean(log_return_20d)`	21
16	`day_of_week_sin`	`sin(2*pi*weekday/7)`	1
17	`day_of_week_cos`	`cos(2*pi*weekday/7)`	1
18	`month_sin`	`sin(2*pi*(month-1)/12)`	1
19	`month_cos`	`cos(2*pi*(month-1)/12)`	1

C Run configuration

Table 22: The committed run configuration, verbatim from the bundle. These values enter the configuration hash, so a change to any of them produces a different run identity rather than a silently different result.

Parameter	Value
Scope	
`experiment_scope`	`fixed_bist_large_cap_prototype`
`methodology_version`	`accepted-baseline-v2`
`seed`	`42`
Fold geometry	
`min_train_dates`	`24`
`validation_dates`	`10`
`step_dates`	`10`
`embargo_dates`	`1`
Portfolio	
`portfolio_model`	`ridge`
`top_k`	`3`
`starting_equity`	`100000`
`min_trade_value`	`100`
`max_participation`	`0.01`
`liquidity_lookback_sessions`	`20`
`decision_cost_rate`	`0.0001`
Cost schedule	
`commission_rate`	`0.0002`
`bid_ask_spread_rate`	`0.001`
`slippage_rate`	`0.0003`
`market_impact_coefficient`	`0.0001`
`tax_rate`	`0`
Resampling and search	
`bootstrap_iterations`	`10000`
`bootstrap_block_sizes`	`1,2,3,5,8,13,21`
`sensitivity_grid`	`train=24,36,48|val=5,10,20|embargo=1,2|topk=1,2,3,4`

D Executed folds

Table 23: The partition that was executed, read from the run's fold artifact rather than recomputed for the manuscript. Training windows expand from a common start; the embargo column counts trading dates removed between training and validation; validation blocks are disjoint, so no session is evaluated twice.

Fold	Train from	Train to	Dates	Emb.	Validate from	Validate to	Dates
fold_0001	2025-09-01	2025-10-02	24	1	2025-10-06	2025-10-17	10
fold_0002	2025-09-01	2025-10-16	34	1	2025-10-20	2025-11-03	10
fold_0003	2025-09-01	2025-10-31	44	1	2025-11-04	2025-11-17	10
fold_0004	2025-09-01	2025-11-14	54	1	2025-11-18	2025-12-01	10
fold_0005	2025-09-01	2025-11-28	64	1	2025-12-02	2025-12-15	10
fold_0006	2025-09-01	2025-12-12	74	1	2025-12-16	2025-12-29	10
fold_0007	2025-09-01	2025-12-26	84	1	2025-12-30	2026-01-13	10
fold_0008	2025-09-01	2026-01-12	94	1	2026-01-14	2026-01-27	10
fold_0009	2025-09-01	2026-01-26	104	1	2026-01-28	2026-02-10	10
fold_0010	2025-09-01	2026-02-09	114	1	2026-02-11	2026-02-24	10
fold_0011	2025-09-01	2026-02-23	124	1	2026-02-25	2026-03-10	10
fold_0012	2025-09-01	2026-03-09	134	1	2026-03-11	2026-03-25	10

E Full configuration grid

Table 24: Every configuration in the search grid, ordered by per-session Sharpe ratio. Each row is a complete re-run of the evaluation under one fold geometry and one portfolio breadth, not a re-weighting of a cached result. The final column names the model with the best out-of-sample R_0^2 in that trial.

Configuration	Folds	Sessions	Net return	Sharpe	Round trips	Best model
train24_val5_step5_emb2_k3	25	125	4.63%	0.0356	155	zero_return
train24_val5_step5_emb2_k1	25	125	4.28%	0.0304	67	zero_return
train24_val10_step10_emb2_k1	12	120	1.67%	0.0169	65	zero_return
train24_val5_step5_emb2_k2	25	125	1.62%	0.0163	115	zero_return
train24_val5_step5_emb2_k4	25	125	-0.19%	0.0049	189	zero_return
train36_val20_step20_emb1_k1	5	100	-0.67%	0.0013	49	zero_return
train36_val20_step20_emb2_k1	5	100	-1.54%	-0.0054	49	zero_return
train24_val10_step10_emb2_k3	12	120	-1.69%	-0.0060	150	zero_return
train24_val5_step5_emb1_k3	25	125	-2.36%	-0.0092	146	zero_return
train48_val20_step20_emb2_k1	5	100	-2.68%	-0.0102	69	zero_return
train48_val10_step10_emb2_k2	10	100	-2.53%	-0.0113	122	zero_return
train48_val20_step20_emb2_k2	5	100	-3.05%	-0.0161	131	zero_return
train24_val10_step10_emb1_k1	12	120	-3.58%	-0.0162	62	zero_return
train24_val5_step5_emb1_k2	25	125	-4.54%	-0.0211	109	zero_return
train48_val20_step20_emb1_k1	5	100	-5.16%	-0.0282	68	zero_return
train24_val10_step10_emb1_k3	12	120	-4.77%	-0.0284	143	zero_return
train36_val10_step10_emb2_k2	11	110	-5.41%	-0.0315	121	zero_return
train24_val5_step5_emb1_k1	25	125	-6.99%	-0.0343	64	zero_return
train24_val20_step20_emb2_k1	6	120	-7.36%	-0.0371	73	zero_return
train24_val5_step5_emb1_k4	25	125	-6.52%	-0.0372	180	zero_return
train24_val10_step10_emb2_k2	12	120	-6.57%	-0.0380	111	zero_return
train24_val10_step10_emb2_k4	12	120	-6.14%	-0.0388	183	zero_return
train36_val10_step10_emb2_k1	11	110	-8.37%	-0.0488	66	zero_return
train24_val20_step20_emb2_k3	6	120	-8.20%	-0.0492	178	zero_return
train24_val20_step20_emb1_k1	6	120	-9.57%	-0.0507	71	zero_return
train48_val20_step20_emb1_k2	5	100	-7.28%	-0.0512	129	zero_return
train36_val20_step20_emb1_k2	5	100	-7.10%	-0.0528	85	zero_return
train48_val10_step10_emb2_k1	10	100	-8.86%	-0.0529	65	zero_return
train24_val10_step10_emb1_k4	12	120	-8.59%	-0.0575	174	zero_return
train48_val5_step5_emb2_k2	20	100	-8.49%	-0.0577	115	zero_return
train48_val10_step10_emb1_k2	10	100	-8.35%	-0.0579	118	zero_return
train36_val20_step20_emb2_k2	5	100	-7.77%	-0.0586	86	zero_return
train36_val10_step10_emb1_k2	11	110	-8.96%	-0.0591	121	zero_return

Configuration	Folds	Sessions	Net return	Sharpe	Round trips	Best model
train36_val10_step10_emb1_k1	11	110	-9.82%	-0.0592	65	zero_return
train24_val10_step10_emb1_k2	12	120	-9.62%	-0.0599	106	zero_return
train24_val20_step20_emb2_k2	6	120	-10.81%	-0.0667	128	zero_return
train48_val10_step10_emb1_k1	10	100	-10.73%	-0.0697	62	zero_return
train48_val20_step20_emb2_k3	5	100	-10.10%	-0.0713	187	zero_return
train24_val20_step20_emb1_k3	6	120	-11.29%	-0.0718	172	zero_return
train48_val5_step5_emb2_k3	20	100	-9.74%	-0.0723	153	zero_return
train36_val5_step5_emb2_k2	22	110	-11.47%	-0.0744	111	zero_return
train24_val20_step20_emb2_k4	6	120	-12.57%	-0.0829	222	zero_return
train36_val10_step10_emb1_k3	11	110	-11.47%	-0.0843	170	zero_return
train36_val20_step20_emb1_k3	5	100	-9.90%	-0.0864	114	zero_return
train36_val5_step5_emb1_k2	22	110	-13.05%	-0.0871	108	zero_return
train36_val5_step5_emb2_k1	22	110	-14.34%	-0.0900	62	zero_return
train24_val20_step20_emb1_k2	6	120	-14.07%	-0.0914	124	zero_return
train36_val10_step10_emb2_k3	11	110	-12.22%	-0.0916	168	zero_return
train36_val20_step20_emb2_k3	5	100	-10.70%	-0.0945	116	zero_return
train48_val20_step20_emb1_k3	5	100	-12.64%	-0.0956	184	zero_return
train24_val20_step20_emb1_k4	6	120	-14.96%	-0.1016	214	zero_return
train48_val10_step10_emb2_k3	10	100	-13.98%	-0.1075	167	zero_return
train48_val5_step5_emb2_k4	20	100	-13.87%	-0.1086	186	zero_return
train48_val5_step5_emb2_k1	20	100	-15.86%	-0.1097	64	zero_return
train36_val5_step5_emb2_k3	22	110	-15.51%	-0.1152	153	zero_return
train36_val5_step5_emb1_k3	22	110	-16.25%	-0.1213	150	zero_return
train48_val20_step20_emb2_k4	5	100	-15.38%	-0.1226	238	zero_return
train36_val5_step5_emb1_k1	22	110	-19.14%	-0.1241	60	zero_return
train48_val5_step5_emb1_k1	20	100	-17.47%	-0.1266	61	zero_return
train36_val10_step10_emb1_k4	11	110	-17.02%	-0.1317	212	zero_return
train36_val20_step20_emb1_k4	5	100	-14.34%	-0.1353	141	zero_return
train48_val5_step5_emb1_k2	20	100	-18.46%	-0.1407	110	zero_return
train48_val10_step10_emb2_k4	10	100	-17.83%	-0.1430	207	zero_return
train36_val20_step20_emb2_k4	5	100	-15.04%	-0.1433	143	zero_return
train48_val20_step20_emb1_k4	5	100	-17.33%	-0.1436	234	zero_return
train48_val10_step10_emb1_k3	10	100	-16.85%	-0.1438	163	zero_return
train36_val10_step10_emb2_k4	11	110	-18.22%	-0.1445	209	zero_return
train48_val5_step5_emb1_k3	20	100	-17.77%	-0.1449	149	zero_return
train36_val5_step5_emb2_k4	22	110	-19.32%	-0.1501	184	zero_return
train36_val5_step5_emb1_k4	22	110	-20.13%	-0.1576	180	zero_return
train48_val5_step5_emb1_k4	20	100	-21.55%	-0.1829	181	zero_return
train48_val10_step10_emb1_k4	10	100	-20.98%	-0.1852	202	zero_return

F Mutation catalogue

Table 25: The 19 defects the mutation harness reintroduces into the source. For each one it runs the guarding test clean, applies the edit, requires the test to fail, restores the file, and requires the test to pass again. A defect that survives means the guarding test is decorative; three were found that way and repaired.

#	Subject	Defect reintroduced
1	variance guard	compare dispersion against zero instead of the series scale
2	variance guard (sharpe module)	same defect on the per-period estimator
3	holding period	hardcode one session instead of reading the ledger
4	Holm	drop the running maximum that enforces monotonicity
5	Diebold-Mariano	drop the Harvey-Leybourne-Newbold small-sample correction
6	Diebold-Mariano	aggregate rows instead of sessions, ignoring cross-sectional dependence
7	Diebold-Mariano	guard the differential variance against zero, not against scale
8	HAC	normalise the autocovariance by n-k instead of n
9	SPA	compare floored maxima, collapsing the comparison onto the atom at zero
10	SPA	skip the bootstrap recentring entirely
11	Lo annualisation	sum q-1 autocorrelations from a 120-session sample
12	Deflated Sharpe	ignore the trial count and never deflate
13	Cross-sectional dependence	assume independence by ignoring the correlation
14	Stationary bootstrap	use fixed-length blocks instead of geometric ones
15	Sensitivity grid	decouple the step from the fold width
16	Accepted config	allow a grid that omits the reported configuration
17	Clark-West	drop the factor of two from the encompassing adjustment
18	Clark-West	make the one-sided test two-sided and halve its power
19	Joint search	resample each configuration on its own index draw